Chapter 9

Credit Risk Modeling

9.1 Introduction

Chapter 8 developed the standard toolkit for market risk: Value at Risk, Expected Shortfall, and the volatility and correlation machinery behind them. This chapter turns to a related but distinct problem: credit risk, the risk that a counterparty (a bond issuer, a loan borrower, or a derivatives counterparty) fails to meet a contractual obligation. Where market risk asks how much a position's *value* could move, credit risk asks whether a counterparty will *pay at all*. That is a binary event with its own probability, its own severity, and its own regulatory and modeling apparatus: probability of default, loss given default, credit scoring, structural models, and rating systems.

The chapter proceeds from the single borrower to the whole portfolio. It begins with the three-component decomposition of expected loss and what drives each component. It then covers three distinct ways of estimating a borrower's probability of default (a statistical model fit to historical data, a structural model that reads default risk out of equity prices, and a market-implied approach that reads it out of credit derivative spreads) and extends single-borrower ratings into multi-year transition matrices. Along the way it treats climate change as a credit stress scenario and, once portfolio correlation is in place, the credit risk of derivatives, whose exposure is itself random. It closes with the portfolio, where the correlation between borrowers' defaults, not just their individual probabilities, drives extreme loss, illustrated by a worked single-factor model and by a historical episode in which underestimating that correlation contributed to a systemic financial crisis.

One worked example runs through much of the chapter: a small loan portfolio that reuses the credit-risk-adjusted bond from Chapter 5 and the loan dataset from Chapter 6.

9.2 Measuring Credit Risk

Credit risk decomposes into a probability, an exposure, and a loss given that it happens; this part builds each and scores a borrower.

9.2.1 Credit Risk Fundamentals: Expected Loss

Suppose a bank holds a bond and wants a single number summarizing how much of that exposure it should expect to lose to default, on average across many similar bonds, before knowing whether this one defaults. Credit risk is conventionally decomposed into three components whose product answers this, giving the expected loss on a credit exposure:

$$\text{Expected Loss} = \text{PD} \times \text{LGD} \times \text{EAD}, \tag{9.1}$$

where PD is the probability of default over the relevant horizon (typically one year), LGD is the loss given default, the fraction of the exposure not recovered after default, and EAD is the exposure at default, the dollar amount outstanding at the time of default (which can differ from the current balance for instruments like credit lines that can be drawn down further before default).

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Recall the corporate bond from Chapter 5's discussion of pricing default risk: face value \$1,000, one-year default probability PD = 3%, and a recovery rate of 40% of face value, so LGD = 1 − 0.40 = 60%. Treating the full face value as the exposure at default, EAD = \$1,000, the expected loss is

$$\text{Expected Loss} = 0.03 \times 0.60 \times 1{,}000 = \$18.$$

Figure 9.1 lays out the decomposition as a pipeline; the rest of the chapter estimates each input in turn, first PD (via logistic regression and, later, the Merton model), then LGD, then EAD.

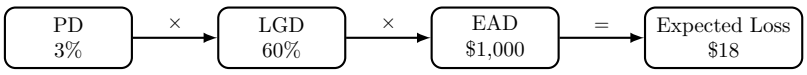


Figure 9.1: The expected-loss decomposition, applied to the Chapter 5 bond example

This matches Chapter 5's earlier calculation: the bond's expected payoff was \$982 against a promised \$1,000, and 1,000 − 982 = 18, so discounting the expected payoff and computing an expected loss are two equivalent descriptions of the same credit risk.

The two framings differ in purpose rather than substance. The pricing framing asks "what should this bond be worth today." The expected-loss framing asks "how much should a lender expect to lose, on average, and how much capital or loan-loss reserve should be held against that possibility," the accounting and regulatory question addressed later in this chapter.

EAD is the least discussed of the three components, but it deserves its own worked example, since it is not simply "the current balance" whenever a borrower has undrawn credit. Consider a committed revolving credit line with a \$500,000 total commitment, currently drawn to \$200,000, leaving \$300,000 undrawn. Many borrowers draw down a credit line further as their condition deteriorates, often because they are approaching default and want to secure cash while they can, so exposure at default must include some fraction of the undrawn commitment.

Basel's standardized approach captures this with a credit conversion factor (CCF), the assumed fraction of the undrawn commitment that converts to drawn exposure by the time of default; for an unconditionally cancellable commercial credit line, a CCF of 50% is a representative assumption, giving

$$\text{EAD} = \text{Drawn} + \text{CCF} \times \text{Undrawn} = 200{,}000 + 0.50(300{,}000) = \$350{,}000,$$

75% higher than the \$200,000 currently drawn. That is why credit lines and other commitment-based products carry more expected loss, for a given PD and LGD, than a fully drawn term loan of the same balance. Banks sophisticated enough to model EAD directly, rather than relying on a regulatory CCF table, estimate it from their own history of drawn-versus-committed balances at past defaults.

9.2.2 What Drives Loss Given Default

PD receives the most modeling attention, since it is most directly tied to the statistical and machine learning methods of Chapter 6, but LGD is just as consequential for expected loss and harder to model, since recovery outcomes depend on factors observed only after a default has occurred.

Seniority and collateral are the two most important drivers. A secured loan backed by specific collateral (real estate, equipment, inventory) typically recovers far more in default than an unsecured loan to the same borrower, since the lender has a direct claim on identifiable assets rather than competing with every other unsecured creditor. Seniority matters similarly: senior debt is repaid before subordinated debt in a bankruptcy, so a subordinated bondholder can recover substantially less than a senior bondholder of the same defaulted issuer, despite facing the same probability of default. Recovery rates are also cyclical in a way that compounds portfolio credit risk. In a recession, defaults become more frequent (PD rises) at the same time that collateral values fall and distressed-asset markets fill with sellers, so LGD rises too. PD and LGD are therefore positively correlated across the credit cycle, not independent as this chapter's simplified expected-loss and economic-capital formulas assume for tractability. A lender that estimates LGD as a full-cycle average, rather than conditioning it on current or forecast conditions, will understate losses during the downturns when accurate estimates matter most.

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Collateral coverage, the ratio of collateral value to loan balance, is estimable from a lender's own default history, and a simple linear regression illustrates the pattern numerically. Consider four hypothetical defaulted loans:

$$\begin{aligned} &\text{Coverage} = 0.40, \quad \text{Recovery} = 25\%, \\ &\text{Coverage} = 0.60, \quad \text{Recovery} = 45\%, \\ &\text{Coverage} = 0.80, \quad \text{Recovery} = 55\%, \\ &\text{Coverage} = 1.20, \quad \text{Recovery} = 75\%. \end{aligned}$$

Fitting $\text{Recovery} = a + b\,(\text{Coverage})$ by ordinary least squares,

$$\hat{b} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = 0.60, \qquad \hat{a} = \bar{y} - \hat{b}\bar{x} = 0.05,$$

so $\widehat{\text{Recovery}} = 0.05 + 0.60\,(\text{Coverage})$, with $R^2 \approx 0.969$, a strong linear fit given only four points.

A loan with coverage exactly at 1.0 (collateral value equal to the outstanding balance) has a predicted recovery of $0.05 + 0.60(1.0) = 65\%$, implying LGD = 35%. That is meaningfully lower than the 60% LGD assumed for this chapter's unsecured bond: the collateral effect made numerical. A lender with enough default history can estimate the relationship directly, applying regression to the loss side of credit risk rather than the default-probability side.

The high R^2 deserves the caution Chapter 6 raised about small samples, and the slope's standard error, computed as for that chapter's credit-spread regression, makes it concrete. With only two degrees of freedom (four observations, two estimated parameters), $\text{SE}(\hat{b}) \approx 0.0756$, giving a t-statistic of $0.60/0.0756 \approx 7.94$ and a 95% confidence interval for the slope of roughly $(0.27, 0.93)$. The slope is still significant at conventional levels ($p \approx 0.016$), but the confidence interval is enormous relative to the point estimate: any true slope from 0.27 to 0.93 is consistent with the data. Propagated through, the predicted recovery at a coverage ratio of 1.0 could plausibly range from roughly 32% to 98%, an enormous range for a single reported LGD figure. A lender relying on a regression estimated from a handful of defaults should therefore treat the fitted line as a starting point, and collect more default history or borrow estimates from a broader industry dataset before using it for capital or pricing.

9.2.3 Credit Stress Testing: Stressed PD and LGD

Credit stress testing makes this procyclicality concrete. Rather than assuming the base-case PD and LGD estimated from calm, average conditions, a stress test recomputes expected loss under a specified adverse scenario (a recession, a sector downturn, a sharp decline in collateral values) and asks how much larger it becomes.

Applying a stress scenario to the same bond from Section 9.2.1, raising the default probability from 3% to a stressed PD = 8% (about 2.7 times its base-case value) and raising loss given default to a stressed LGD = 70% (reflecting weaker collateral values in a downturn, the mechanism Section 9.2.2 described), the stressed expected loss is

$$\text{Expected Loss}^{\text{stress}} = 0.08 \times 0.70 \times 1{,}000 = \$56,$$

more than triple the base-case expected loss of \$18, although neither input moved implausibly on its own. That is why treating PD and LGD as independent, average-cycle constants can mislead capital planning: a lender reserving only against calm-period expected loss will be under-reserved in a recession, when the shortfall matters most. Regulatory exercises such as the Federal Reserve's CCAR, introduced from the market-risk side in Chapter 8, apply the same stressed-PD-and-LGD logic across a bank's entire loan book under a specified adverse macroeconomic scenario.

9.2.4 Climate Risk in Credit Portfolios

Climate change is the stress scenario that supervisors have most actively added to that list over the past decade, and it reaches a loan book through two distinct channels. *Physical risk* is the damage done by the climate itself: acute events such as floods, hurricanes, and wildfires, and chronic shifts such as rising sea levels

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and heat stress. It destroys collateral (raising LGD) and interrupts borrowers' revenue (raising PD), both in geographic clusters, so the losses arrive together rather than diversifying. *Transition risk* is the damage done by the move to a low-carbon economy: carbon prices, emissions regulation, cheaper clean technology, and shifting customer preferences erode the profitability of carbon-intensive borrowers, and they can *strand* the assets that secure their loans, oil reserves that will never be extracted and coal plants retired before the end of their economic life. Neither channel is a new kind of risk in the sense of this chapter. Both arrive as changes in PD, LGD, and EAD, which is why supervisors treat climate as a *driver* of existing credit, market, and operational risks rather than as a fourth category with its own capital charge.

What makes climate risk hard is that the statistical approach of the rest of this chapter has nothing to estimate from. The horizon runs to decades, and no historical sample contains an economy-wide carbon price at the levels policy scenarios contemplate, so no default history can say how borrowers respond. Supervisors have therefore built the discipline around scenario analysis. The Network for Greening the Financial System (NGFS), a network of central banks and supervisors, publishes reference scenarios grouped into orderly paths such as Net Zero 2050, disorderly paths such as Delayed Transition, and "hot house world" paths such as Current Policies, in which transition risk stays low and physical risk grows. Its fifth vintage, released in November 2024, estimated that a carbon price of around $300 per tonne of CO_2 would be needed by 2035 to put the world on a net-zero path by 2050 (NGFS 2024). The Bank of England's 2021 Climate Biennial Exploratory Scenario ran UK banks and insurers through three 30-year scenarios and published its results in May 2022 (Bank of England 2022). The European Central Bank's 2022 climate risk stress test covered 104 banks, and the 41 that ran its detailed bottom-up module projected about €70 billion of combined credit and market losses, a figure the ECB itself described as a substantial understatement (ECB 2022). A joint ECB and European Supervisory Authorities "Fit-for-55" scenario analysis followed in November 2024. In the United States, the Federal Reserve ran a pilot exercise with six large banks in 2023 and published a summary in May 2024 with no capital consequences (Board of Governors of the Federal Reserve System 2024), and in October 2025 the federal banking agencies withdrew their interagency climate risk-management principles for large banks altogether (Board of Governors of the Federal Reserve System et al. 2025). In Canada, OSFI's Guideline B-15 on climate risk management took effect for the largest banks from fiscal year-end 2024 (OSFI 2023). As of October 2026, then, climate scenario analysis is a firm supervisory expectation in Europe, the United Kingdom, and Canada, and a matter left to the general safety-and-soundness standards in the United States. None of these exercises sets capital directly; they are exploratory, reflecting how immature the modeling still is.

The modeling problem has a standard shape. A scenario supplies a path of macroeconomic and policy variables (a carbon price, energy prices, GDP by sector); a *transmission model* translates that path into each borrower's financial statements; and the credit models of this chapter translate the statements into PD and LGD. The judgment lives in the transmission step. A simple and transparent version measures the carbon cost a borrower would bear as a share of its earnings,

$$c = \frac{e \times p \times (1 - \pi)}{m}, \qquad \mathrm{PD}^{\mathrm{climate}} = N\big(N^{-1}(\mathrm{PD}) + \kappa\, c\big)\,, \tag{9.2}$$

where e is the borrower's emissions intensity in tonnes of CO_2 per dollar of revenue, p the carbon price, π the fraction of the cost the borrower can pass on to its customers, and m its EBITDA margin, so that c is the carbon bill as a fraction of EBITDA. The second expression shifts the borrower's default threshold by κc standard deviations, the probit scale on which the single-factor model of Section 9.4.2 moves a default threshold; κ is the threshold's sensitivity to an earnings shock, which a bank would calibrate from a Merton-style model of each sector's leverage or from its own history. Machine learning enters mainly at the data end: emissions are reported for only a minority of corporate borrowers, so they are imputed from sector, revenue, and asset data, and physical-risk exposure is mapped from satellite imagery and geocoded collateral, the methods of Chapter 17 and of the catastrophe models in Chapter 11.

9.2.5 Worked Example 9.1: A Carbon-Price Shock to a Sector Loan Book

Consider a $1 billion corporate loan book spread over four sectors, with the base-case PDs and LGDs in Table 9.1, and apply a stylized, immediate carbon price of $p = \$100$ per tonne with no pass-through ($\pi = 0$) and a threshold sensitivity of $\kappa = 2$, so that a carbon bill equal to half of EBITDA moves a borrower's default

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threshold by one full standard deviation. These are stated modeling assumptions, not estimates. For the fossil-heavy utilities, which emit 1,500 tonnes per million dollars of revenue on a 30% margin, the carbon bill is $1{,}500 \times 100/1{,}000{,}000 = 15\%$ of revenue, or $c = 0.15/0.30 = 50\%$ of EBITDA, and Equation (9.2) gives

$$\mathrm{PD}^{\mathrm{climate}} = N\big(N^{-1}(0.010) + 2 \times 0.50\big) = N(-2.326 + 1.000) = N(-1.326) = 9.24\%.$$

Sector	EAD	PD	LGD	c	$\mathrm{PD}^{\mathrm{climate}}$	EL	$\mathrm{EL}^{\mathrm{climate}}$
Oil and gas	200	2.00%	45%	30.00%	7.30%	1.800	6.571
Fossil-heavy utilities	250	1.00%	40%	50.00%	9.24%	1.000	9.236
Transportation	150	2.50%	50%	25.00%	7.21%	1.875	5.411
Commercial real estate	400	1.50%	35%	1.33%	1.60%	2.100	2.246
Total	1,000					6.775	23.464

Table 9.1: A \$100-per-tonne carbon-price shock to a four-sector loan book, with EAD and expected loss in \$ millions. Emissions intensities are 600, 1,500, 300, and 20 tonnes per \$1 million of revenue and EBITDA margins 20%, 30%, 12%, and 15%; c is the carbon bill as a share of EBITDA.

Applying EL = PD × LGD × EAD sector by sector, the book's expected loss rises from \$6.775 million to \$23.464 million, a factor of 3.46, comparable in size to Section 9.2.3's recession scenario but with an entirely different footprint. The utilities sector, the *safest* in the book on base-case PD, becomes the largest single source of stressed loss, \$9.236 million, and utilities and oil and gas together account for 77.9% of the \$16.689 million increase, while the commercial real estate book, the largest exposure, barely moves. A base-case rating says almost nothing about transition vulnerability, which is why supervisors ask for sector- and borrower-level analysis rather than a book-wide PD multiplier. Pass-through matters as much as the carbon price: if borrowers can pass half the cost on to customers, $\pi = 0.5$, stressed expected loss falls to \$12.411 million, so a single behavioral parameter that no historical data pins down roughly halves the answer.

The example is deliberately crude, and its caveats are the substance of climate credit modeling. The mapping in Equation (9.2) is static: real borrowers abate emissions, change their product mix, and invest in cleaner capacity as the carbon price rises, so a fixed intensity e overstates the long-run hit to the firms best able to adapt and understates it for those that cannot. The sensitivity κ has no data from which to estimate it in the relevant range. The emissions themselves are often imputed rather than reported, which turns a data-quality problem into a model-risk problem, and an imputed figure for a single large borrower can move the whole sector's result. Even the scenarios carry model risk: the NGFS noted in 2025 that a research paper underpinning part of its fifth vintage's physical-damage estimates had since been retracted (NGFS 2024). A climate stress result is therefore best read as conditional on a long chain of stated assumptions, and most useful for the ranking it produces (which sectors and borrowers are most exposed) rather than the headline dollar figure.

9.2.6 Credit Scoring with Logistic Regression

Expected loss requires a PD for each borrower, and Chapter 6 introduced the standard tool for producing one: logistic regression, which models the probability of a binary outcome (here, default within the horizon) as a function of borrower characteristics. Recall the small loan dataset from Chapter 6's K-nearest neighbors example, with six applicants described by debt-to-income ratio and years of credit history. Fitting a logistic regression to that same dataset,

$$\Pr(\text{Default}) = \frac{1}{1 + e^{-z}}, \qquad z = -12.152 + 0.530\,(\text{Debt/Income}) - 0.231\,(\text{Credit History}),$$

gives coefficients that make economic sense: higher leverage increases the log-odds of default (positive coefficient), while a longer credit history reduces it (negative coefficient). These coefficients come from scikit-learn's default fit, which adds an L2 (ridge) penalty with $C = 1$, and on this dataset the penalty is not a detail. Debt-to-income alone separates the two classes perfectly (the three defaulters sit at 30–40%, the

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three non-defaulters at 15–20%), and under such *complete separation* the unpenalized maximum-likelihood estimate does not exist. The likelihood keeps rising as the coefficients grow without bound, so a plain fit (statsmodels, for example) either fails with a perfect-separation error or returns arbitrarily large coefficients. The quoted values reflect the penalty as much as the data, which is why real scorecards on small or cleanly split samples use penalized or Firth-corrected logistic regression, and why this sample's perfect in-sample AUC later in the chapter is a warning sign rather than an achievement. Applying this to the same new applicant considered in Chapter 6 (debt-to-income of 28%, 4 years of credit history),

$$z = -12.152 + 0.530(28) - 0.231(4) \approx 1.761, \qquad \Pr(\text{Default}) = \frac{1}{1 + e^{-1.761}} \approx 85.3\%.$$

This is consistent with, and more precise than, the KNN classification of Chapter 6, which produced only a discrete "predicted default" label from a 2-to-1 majority vote among the three nearest neighbors. Logistic regression reports a continuous probability, the PD input the expected-loss formula requires.

Lenders often rescale a model's probability into a more interpretable credit score with a linear transformation, for example

$$\text{Score} = 850 - 550 \times \Pr(\text{Default}),$$

chosen so that a default probability near 0 produces a score near 850 (the top of a typical consumer credit-score scale) and a default probability near 1 produces a score near 300 (the bottom). For the applicant above, $\text{Score} = 850 - 550(0.853) \approx 381$, a score that would place this applicant in a high-risk tier under most lenders' underwriting policies. The scaling constants are a business choice, not a statistical one, but the logic of mapping a probability onto a scale loan officers and borrowers can interpret is universal. It connects to the calibration discussion of Chapter 6: a credit score is meaningful only if the underlying probability is accurately calibrated, not merely useful for ranking applicants.

A subtler difficulty affects every credit scoring model built this way, however well it is fit or calibrated. A lender observes loan performance only for applicants it approved, since a rejected applicant never receives a loan and never generates an outcome. A model trained only on approved applicants, like the six-applicant sample here, never sees how someone the previous policy judged too risky would have performed, a selection bias that can silently distort the fitted relationship between borrower characteristics and default risk. Reject inference is the general name for techniques that try to correct this, from simple approaches (assigning rejected applicants an assumed outcome based on their characteristics) to statistical corrections that model the approval decision alongside the default outcome. None is fully satisfying, since the counterfactual (what would have happened had a rejected applicant been approved) is unobservable. Reject inference therefore remains an imperfect compromise, and a model's performance on its training population can differ meaningfully from its performance on the full population it screens. It is a credit-specific instance of a general lesson: data generated by a decision process, here a prior lending policy's approvals and rejections, is never a neutral sample of the population that process was applied to.

9.2.7 Validating a Credit Scoring Model: The Gini Coefficient and AUC

Ranking applicants correctly from safest to riskiest is a distinct property from calibration, and it has its own standard validation metric in credit scoring: the area under the ROC curve (AUC), or equivalently the Gini coefficient, $\text{Gini} = 2\,\text{AUC} - 1$, which rescales AUC so that a purely random ranking scores 0 and a perfect ranking scores 1. Both answer the question introduced in Chapter 6: across every pair of one defaulter and one non-defaulter, how often does the model assign the defaulter a strictly higher predicted default probability?

Applying the fitted logistic regression above to all six applicants in Chapter 6's Table 6.5, sorted by predicted default probability, gives the values in Table 9.2.

Every one of the nine possible (defaulter, non-defaulter) pairs, three defaulters times three non-defaulters, is correctly ordered: each of C, A, and E's predicted probabilities exceeds each of B, F, and D's, so $\text{AUC} = 9/9 = 1.0$ and $\text{Gini} = 2(1.0) - 1 = 1.0$, a perfect ranking. This is a red flag, not a cause for celebration. The AUC is computed on the same six observations the model was fit to, with two features, so a perfect in-sample ranking is the least surprising outcome imaginable, not evidence the model would rank a new applicant accurately. A credit model's AUC and Gini coefficient are meaningful only on a held-out validation

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Applicant	Debt/income (%)	Pr(Default)	Actual outcome
C	40	0.9998	Default
A	35	0.9974	Default
E	30	0.9550	Default
B	20	0.0323	No default
F	18	0.0144	No default
D	15	0.0015	No default

Table 9.2: Predicted default probabilities for all six training applicants, sorted from highest to lowest risk

sample, following the train-test discipline of Chapter 6, and a near-perfect AUC on a tiny training sample signals overfitting rather than a property to report in a validation document.

9.2.8 Scorecards: Weight of Evidence and Points

The regression above is the right model but not the form a lender deploys. Almost every production consumer credit model is a *scorecard*: the same logistic regression, with each characteristic cut into bins, each bin replaced by a number summarizing its risk, and the result expressed as whole points. Binning lets a linear model capture a nonlinear relationship, makes the model robust to outliers and data errors, and leaves every part of it inspectable row by row.

The number replacing each bin is its *weight of evidence*, the log ratio of the share of non-defaulters to the share of defaulters falling in it,

$$\mathrm{WOE}_i = \ln\!\left(\frac{g_i/G}{b_i/B}\right), \tag{9.3}$$

with g_i, b_i the good and bad counts in bin i and G, B the portfolio totals. Table 9.3 works this for a portfolio of 1,000 loans, 900 performing and 100 defaulted.

Debt/income	Good	Bad	g_i/G	b_i/B	WOE	IV contribution
$< 15\%$	270	10	0.300	0.100	$+1.099$	0.220
15–25%	315	20	0.350	0.200	$+0.560$	0.084
25–35%	225	30	0.250	0.300	-0.182	0.009
$\geq 35\%$	90	40	0.100	0.400	-1.386	0.416
Total	900	100	1.000	1.000	—	0.729

Table 9.3: Weight of evidence and information value for a debt-to-income characteristic

The column is *monotone*, falling as leverage rises; a reversal is read as evidence that the bins are wrong or the sample too thin, and bins are merged until it disappears. The whole characteristic then summarizes into its *information value*, $\mathrm{IV} = \sum_i (g_i/G - b_i/B)\,\mathrm{WOE}_i = 0.729$, which by convention (below 0.02 useless, 0.1–0.3 medium, 0.3–0.5 strong) makes this a strong predictor; values far above 0.5 usually mean the characteristic partly restates the outcome, the credit version of the leakage problem Chapter 6 warns about.

The fitted log-odds are then rescaled so that a fixed number of points always means the same thing. Choosing PDO $= 20$ *points to double the odds* fixes the slope at $20/\ln 2 = 28.854$, and the gap between the safest and riskiest bands becomes $28.854(1.099 + 1.386) = 71.7$ points, or $71.7/20 = 3.59$ doublings, an odds ratio of $2^{3.59} = 12.0$. That checks straight off the table: the odds of being good are $270/10 = 27$ in the lowest band and $90/40 = 2.25$ in the highest, and $27/2.25 = 12.0$ exactly. The points scale adds no information; it repackages the log-odds into units a human can add up.

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9.2.9 The Scorecard Is a Generalized Additive Model

It is tempting to file the scorecard under "the classical method machine learning came to replace." That reading is wrong twice, and correcting it locates where machine learning earns its keep in credit.

Start with the bins. Choosing cut points by hand is possible but not what anyone does; the standard tool is *supervised discretization*, fitting a decision tree to the single characteristic and taking its splits as bin edges. On a simulated 20,000-loan portfolio, a depth-limited tree with a floor of 800 accounts per leaf selects cuts at 17.1%, 32.7%, 41.3%, 44.5%, and 47.3%, giving six bins whose weights of evidence run +1.054 down to −1.433, monotone with no manual merging and an information value of 0.620. The scorecard's most hand-crafted-looking step is done by a tree learner.

Now the functional form. Writing the scorecard out,

$$\ln \frac{p}{1-p} = \beta_0 + \sum_j \beta_j \,\mathrm{WOE}_j(x_j) = \beta_0 + \sum_j f_j(x_j), \tag{9.4}$$

each characteristic contributes an arbitrary piecewise-constant function of its own value, and the contributions add. That is the definition of a *generalized additive model*. A credit scorecard is a GAM fitted by hand, and has been since the 1950s, decades before the term existed.

This matters because the state of the art in interpretable machine learning, the *Explainable Boosting Machine*, fits this object automatically: it boosts shallow trees on one feature at a time, round-robin, so no interaction can form, and the accumulated trees for feature j become f_j. The companion notebook implements this in about twenty lines and fits it to the same portfolio. Comparing its learned shape function for debt-to-income against the hand-built WOE curve over a 150-point grid gives a correlation of 0.977. They are not analogous; they are two estimates of the same function.

The accuracy comparison is then informative rather than a horse race. On held-out data the scorecard scores an ROC-AUC of 0.8402, the boosted additive model 0.8296, and an unrestricted gradient-boosted model allowed to learn interactions 0.8328, against a ceiling of 0.8429 set by the true data-generating process, which does contain an interaction between leverage and file length. The scorecard lands within 0.003 of the ceiling and nothing beats it. When marginal effects are monotone, bins are well chosen, and interactions are weak, a scorecard is close to optimal and a more flexible model has little left to find. That, rather than institutional inertia, is why it remains the incumbent in regulated lending.

The corollary is the useful part, and the rest of this chapter's machine learning rests on it. ML earns its keep in credit not by fitting a better function of the same characteristics but by doing what a scorecard cannot: building features from data with no natural bins, producing a term structure rather than a single probability (as the lifetime expected-loss discussion later does), and searching a model space rather than fitting one model.

9.3 Modeling Default

This part moves from scoring an applicant to modeling default itself, from a firm's balance sheet, from ratings, and over a loan's whole life.

9.3.1 Structural Credit Models: The Merton Model

Logistic regression estimates PD statistically, from historical data on borrower characteristics and past defaults. A different approach, due to Merton (1974), estimates PD from market prices using the option pricing machinery of Chapter 5: a firm's equity can be modeled as a call option on the firm's assets, with the face value of its debt as the strike price. The analogy is exact. Shareholders have limited liability: if the firm's assets are worth more than its debt at maturity, shareholders receive the difference; if less, shareholders receive nothing and the firm defaults, with bondholders taking the (insufficient) assets. This is the payoff $\max(V_T - D, 0)$ of a European call with the firm's asset value V_T as the underlying and the debt's face value D as the strike, so the Black-Scholes-Merton formula applies, with asset value in place of a stock price and asset volatility in place of σ:

$$E_0 = V_0 N(d_1) - D e^{-rT} N(d_2), \qquad d_1 = \frac{\ln(V_0/D) + (r + \sigma_V^2/2)T}{\sigma_V \sqrt{T}}, \qquad d_2 = d_1 - \sigma_V \sqrt{T}, \tag{9.5}$$

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where E_0 is the market value of equity, V_0 is the (unobserved) market value of the firm's assets, and σ_V is asset volatility. The probability of default, the probability that assets fall below the debt's face value at maturity, is the risk-neutral probability that the option expires out of the money,

$$\Pr(\text{Default}) = N(-d_2),$$

the same $N(-d_2)$ term that appeared when Chapter 5 priced a European put. Figure 9.2 draws the payoff: nothing if assets are worth less than the debt at maturity, the excess above debt otherwise, a call option's hockey stick.

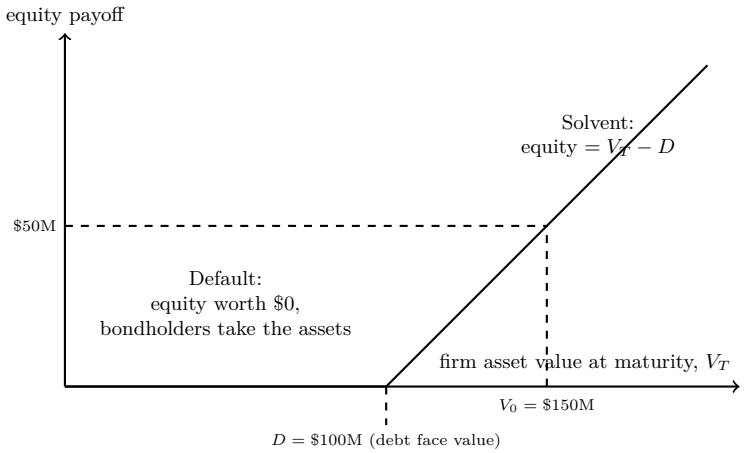


Figure 9.2: The Merton model's equity payoff at maturity, $\max(V_T - D, 0)$, for the worked example's $D = \$100$ million

Consider a firm with

$$V_0 = \$150 \text{ million (estimated asset value)}, \qquad D = \$100 \text{ million (debt due in one year)},$$
$$\sigma_V = 25\% \text{ (asset volatility)}, \qquad r = 4\% \text{ (risk-free rate)}.$$

Then

$$d_1 = \frac{\ln(150/100) + (0.04 + 0.03125)(1)}{0.25} \approx 1.907, \qquad d_2 = 1.907 - 0.25 \approx 1.657,$$

giving an implied equity value of $E_0 = 150\,N(1.907) - 100\,e^{-0.04}N(1.657) \approx \$54.37$ million and a one-year default probability of

$$\Pr(\text{Default}) = N(-1.657) \approx 4.88\%.$$

Unlike logistic regression, the Merton model requires no default history. It needs only an estimate of asset value and volatility, usually inferred from the firm's observable equity value and equity volatility by inverting the same equations. Structural models are therefore especially useful for large public firms with liquid equity and few or no historical defaults, the firms for which a statistical model would have the least data. The commercial descendant of this approach, Moody's KMV model, converts the model's "distance to default" (d_2 above) into a market-implied probability of default that updates in real time as equity prices move, without waiting for a rating agency's periodic review.

9.3.2 Inverting the Model: Recovering Asset Value and Volatility from Equity

The worked example ran the Merton model forward, from asset value and asset volatility to equity value and default probability. In practice neither V_0 nor σ_V is observable (no market quotes a firm's total asset value), so a real application runs the model in reverse, from the firm's equity price back to the unobserved asset value and volatility. This is the KMV approach, made concrete here.

The equity value equation from above, $E_0 = V_0 N(d_1) - De^{-rT}N(d_2)$, is one equation in the two unknowns V_0 and σ_V; a second equation comes from recognizing that equity, as a call option on the firm's assets, has

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its own volatility linked to the asset volatility through the option's delta, $N(d_1)$ (the same Greek introduced in Chapter 5):

$$\sigma_E E_0 = N(d_1)\,\sigma_V V_0. \tag{9.6}$$

Given a firm's observable market equity value E_0 and equity volatility σ_E (estimated from historical or option-implied stock returns, as in Chapters 5 and 7), the two equations can be solved simultaneously for V_0 and σ_V, typically with a numerical root-finder, since d_1 and d_2 depend on V_0 and σ_V nonlinearly.

As a consistency check, apply the inversion to the firm of the forward example. Its equity value was $E_0 \approx$ \$54.37 million; its implied equity volatility, from the same d_1, is $\sigma_E = N(1.907)(0.25)(150)/54.37 \approx 67.03\%$, far above the 25% asset volatility, because a leveraged equity stake is much more volatile than the assets it is a claim on. Feeding only these two observable quantities, with the known debt, rate, and maturity,

$$E_0 = \$54.37 \text{ million}, \qquad \sigma_E = 67.03\%,$$
$$D = \$100 \text{ million}, \qquad r = 4\%, \qquad T = 1 \text{ year},$$

into a numerical solver recovers $\hat{V}_0 = \$150.00$ million and $\hat{\sigma}_V = 25.00\%$ exactly, and therefore the same default probability of 4.88%.

Exact recovery is expected here, since the "observed" equity inputs were generated from the true V_0 and σ_V, but it confirms that the inversion is well posed. A real application feeds in a firm's market equity value and estimated equity volatility, with no knowledge of V_0 or σ_V, and recovers a market-implied estimate of both that updates every time the stock price moves, which makes the KMV approach useful for monitoring large public borrowers between rating reviews.

9.3.3 Market-Implied Default Probabilities: Credit Default Swap Spreads

The Merton model backs out a default probability from a firm's *equity* price. A credit default swap (CDS), introduced in Chapter 3 as a contract that transfers credit risk for a periodic premium, offers a more direct route, since a CDS spread is the market's own price for insuring against default. A standard simplification, sometimes called the credit triangle, relates the CDS spread s directly to a constant hazard rate (default intensity) h and the recovery rate R assumed on the underlying debt,

$$h \approx \frac{s}{1-R}, \tag{9.7}$$

with the cumulative default probability over a horizon T given by $\Pr(\text{Default by } T) = 1 - e^{-hT}$, treating default as a Poisson arrival with constant intensity h, the device behind the survival analysis Chapter 15 used for flagged fraud accounts. A one-year CDS trading at $s = 200$ basis points (2%) on debt with an assumed recovery of $R = 40\%$, as in this chapter's bond example, implies

$$h \approx \frac{0.02}{1-0.40} \approx 3.33\%, \qquad \Pr(\text{Default within one year}) = 1 - e^{-0.0333} \approx 3.28\%,$$

close to the PD = 3% assumed for the bond example, a useful cross-check that market-implied and assumed default probabilities broadly agree.

The constant-hazard assumption extends to any horizon. The cumulative five-year default probability implied by the same $h \approx 3.33\%$ is $1 - e^{-0.0333(5)} \approx 15.35\%$, only about 4.7 times the one-year 3.28% over a five-times-longer horizon. Survival compounds multiplicatively, so each later year's default risk applies only to borrowers still surviving, the same compounding behind the multi-year transition-matrix probabilities later in this chapter. Unlike the Merton model, the CDS-implied approach needs no volatility input, only an observed spread and an assumed recovery rate, but it depends on a reasonably liquid CDS market existing. That holds for large, actively traded corporate and sovereign names but not for the vast majority of small and mid-sized borrowers.

The chapter now has three distinct ways of arriving at a probability of default, each suited to a different population. Logistic regression (Section 9.2.6) requires a history of past borrowers and their outcomes, and it is the natural choice for retail and small-business borrowers with no traded equity or debt; its accuracy is bounded by how representative and recent that sample is. The Merton model requires no default history but needs liquid, exchange-traded equity, which suits it to large public companies and not to a privately held

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small business, the opposite population. The CDS-implied approach needs neither a default history nor an equity price, only a liquid credit derivatives market, which narrows it further to a small set of large corporate and sovereign names. A large bank therefore uses all three where each is available: logistic regression or a machine-learned variant for retail and small business, Merton-style models for large public corporates, and CDS-implied spreads as a real-time check on the most liquid names, because no single approach's data requirements are met across the whole portfolio.

When There Is No Market Price: Private Credit

The fastest-growing corner of corporate lending sits where none of the market-implied routes reach. *Private credit* is lending by non-bank funds directly to companies, typically mid-sized firms owned by private-equity sponsors, that are too large or too risky for a bank loan and too small for the bond market. The loans are negotiated bilaterally or among a small club of lenders, held to maturity, and rarely traded. The International Monetary Fund estimated the market at more than $2.1 trillion of assets and committed capital in 2023, about three-quarters of it in the United States, where its share is approaching that of syndicated loans and high-yield bonds (International Monetary Fund 2024). Its borrowers tend to be smaller and more indebted than those in the public leveraged-loan market, and the IMF reported that after the rate rises of 2022 and 2023 more than a third of them had interest costs exceeding their current earnings.

For a risk modeler the defining feature is the absence of prices. There is no CDS to invert and no traded equity for a Merton model, and the loans themselves do not trade, so a fund values them *mark-to-model*, typically once a quarter, from discounted cash flows, comparable public spreads, and the borrower's reported financials. Three consequences follow. Valuations are stale and partly subjective, and they move less than market prices would: the IMF noted that private-credit marks fell by less than leveraged-loan prices in stress periods despite lower credit quality. Returns computed from those smoothed marks understate volatility and correlation with other assets, which flatters every risk-adjusted performance number built on them. And default itself is partly a choice: lenders protected by maintenance covenants can intervene early, while competition with banks for large deals has loosened covenants, and lenders often prefer amendments, maturity extensions, or interest paid in kind to a recorded default. A PD model trained on recorded defaults then learns the lender's renegotiation policy as much as the borrower's credit quality, the selection problem of Section 9.2.6's reject inference. The tools of this chapter still apply (scoring models on borrower financials, Vasicek-style portfolio capital with an assumed correlation), but every input that elsewhere comes from a market must come from a model, and validation must start by asking what the marks themselves measure.

9.3.4 Rating Models and Transition Matrices

Suppose a bond portfolio manager holds an investment-grade bond and wants to know not just its one-year default probability but how likely it is to be downgraded before that. That is a different risk: a downgrade alone can force some funds to sell and can lower the bond's price long before any payment is missed.

Institutional credit risk (corporate bonds, sovereign debt) is often managed through a rating system, a small number of discrete risk categories (such as AAA, AA, A, BBB, down to Default) that summarize creditworthiness. Ratings are not static: a borrower can be upgraded, downgraded, or default over time, and a rating transition matrix records the probability of moving from each rating to every other rating over a fixed horizon, typically one year. Table 9.4 shows a simplified three-state transition matrix with ratings High and Medium plus an absorbing Default state (once a borrower defaults, it remains in that state, so the Default row is $[0, 0, 1]$).

From \ To	High	Medium	Default
High	0.90	0.08	0.02
Medium	0.10	0.80	0.10
Default	0.00	0.00	1.00

Table 9.4: A simplified one-year rating transition matrix

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The one-year default probability is read directly off the matrix: 2% for a High-rated borrower, 10% for a Medium-rated one. Multi-year default probabilities require treating the transition matrix as a Markov chain and computing its matrix power: the two-year transition matrix is $P^2 = P \times P$, and the entry in row i, column Default gives the cumulative two-year default probability starting from rating i. Carrying out this multiplication,

$$P^2_{\text{High}\to\text{Default}} = 0.90(0.02) + 0.08(0.10) + 0.02(1.00) = 0.046,$$

so a High-rated borrower's cumulative two-year default probability is 4.6%, more than double the one-year figure of 2%, since the borrower can now default either directly or after first migrating to the riskier Medium rating. The same calculation for a Medium-rated borrower gives a two-year default probability of 18.2%, and extending to three years ($P^3 = P^2 \times P$) gives 7.596% for a High-rated borrower. This compounding (cumulative default risk growing faster than linearly with horizon because of downgrade-then-default) is the credit analogue of the growing forecast uncertainty of Chapter 7, and it is why long-dated exposures such as a 10-year bond carry more cumulative default risk than one-year probabilities suggest.

Transition matrices also make credit risk visible before a default occurs. A bondholder whose issuer is downgraded from High to Medium has suffered a real economic loss before any missed payment, since a lower-rated bond trades at a lower price. Marking a credit portfolio to its current ratings, rather than waiting for defaults, is the logic of mark-to-market credit models such as CreditMetrics, which combine a transition matrix like Table 9.4 with bond-pricing formulas to estimate a portfolio's value distribution one year forward, including downgrades that never become defaults. This is a richer picture than the default-only economic capital of Section 9.4.1, which treats every outcome short of default as full recovery of value.

Consider a High-rated, 3-year, \$1,000 face-value bond paying a 5% annual coupon, exactly the yield High-rated bonds command, so it trades at par. One year from now it has two years of cash flows left, and its value depends on its rating then. If it remains High-rated, it is still priced at the 5% yield, giving, by Chapter 5's bond-pricing formula,

$$P_{\text{High}} = \frac{50}{1.05} + \frac{1{,}050}{1.05^2} = \$1{,}000.00,$$

unchanged, since coupon and yield coincide. If downgraded to Medium, the market demands a higher yield of, say, 7%,

$$P_{\text{Medium}} = \frac{50}{1.07} + \frac{1{,}050}{1.07^2} = \$963.84,$$

a loss of \$36.16 with no default. If the bond defaults, the bondholder recovers 40% of face value immediately, \$400.00. Weighting each outcome by Table 9.4's one-year transition probabilities for a High-rated borrower (90%, 8%, 2%) gives an expected one-year-forward value of

$$\mathbb{E}[V] = 0.90(1{,}000.00) + 0.08(963.84) + 0.02(400.00) = \$985.11,$$

with a standard deviation of \$84.16 across the three possible outcomes. The expected value sits below today's \$1,000 par purely from downgrade and default risk, separate from any change in the risk-free rate, and most of the \$84.16 dispersion comes from the rare but severe default outcome, not the likelier but modest Medium markdown: the asymmetric, fat-tailed risk a full CreditMetrics-style analysis captures across many rating states and bonds.

9.3.5 Lifetime Expected Loss: IFRS 9 and CECL

Everything so far computes expected loss over one year. Since 2018 that has not been what a bank reports: IFRS 9, the ninth International Financial Reporting Standard, and its U.S. equivalent CECL, the Current Expected Credit Loss standard, require provisioning against losses expected over an exposure's *remaining life*, which makes credit models the largest modeling estate in most banks. The change answers a crisis-era failure of incurred-loss accounting, under which a bank could not recognize a loss until impairment was objectively evident, so provisions stayed low through the boom and arrived in a rush when capital was scarcest.

IFRS 9 sorts exposures into stages: **Stage 1**, performing, at 12-month expected credit loss; **Stage 2**, having suffered a *significant increase in credit risk*, at *lifetime* loss; **Stage 3**, credit-impaired. CECL drops the staging and requires lifetime loss for everything from day one. Either way a one-year PD no longer suffices: what is needed is a *PD term structure*, the marginal probability of default in each future year.

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The transition matrix introduced above already contains one. Take its Medium-rated borrower and Section 9.2.1's bond, EAD = \$1,000 and LGD = 60%, discounting at 5%. Raising the matrix to the nth power and reading the Medium-to-Default entry gives cumulative default probability by year n; differencing gives the marginals.

Year	Cumulative PD	Marginal PD	Undiscounted loss	Discounted loss
1	0.1000	0.1000	\$60.00	\$57.14
2	0.1820	0.0820	\$49.20	\$44.63
3	0.2502	0.0682	\$40.92	\$35.35
4	0.3078	0.0576	\$34.53	\$28.41
5	0.3570	0.0493	\$29.57	\$23.17
			Lifetime ECL	\$188.70

Table 9.5: Lifetime expected credit loss for a five-year Medium-rated exposure

Year two is checkable by hand: a Medium borrower defaults then either by staying Medium and defaulting, $0.80 \times 0.10 = 0.080$, or by being upgraded first, $0.10 \times 0.02 = 0.002$, giving 0.082 marginal and 0.182 cumulative. The marginals decline not because the borrower improves but because the survivors are a shrinking pool. In Stage 1 the bank provisions $0.10 \times 0.60 \times \$1{,}000/1.05 = \$57.14$; in Stage 2, the full \$188.70. Nothing about the borrower changes at transfer, yet $\$188.70/\$57.14 = 3.30$, and that cliff is what makes the significant-increase test one of the highest-stakes classification problems in the bank.

9.3.6 Learning the Term Structure: A Discrete-Time Hazard Model

That calculation assumes the transition matrix is constant for five years. It is not, and the standard requires a forward-looking, scenario-weighted estimate, which a fixed matrix cannot express. Here the term structure stops being arithmetic and becomes a fitted model. The standard formulation is a *discrete-time hazard*: for each loan, in each year it is still alive, model the conditional probability of defaulting during that year,

$$h(a, m) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 a + \beta_2 a^2 + \beta_3 m)}}, \tag{9.8}$$

with a the loan's age and m a macroeconomic index. The quadratic in age captures *seasoning*, the regularity that defaults are rare just after origination, peak a few years in, then decline. Fitted this way it is an ordinary logistic regression on a loan-year panel, which is why it is the workhorse: every tool from Chapter 6 applies unchanged. It is also the discrete-time counterpart of the survival analysis Chapter 15 develops for flagged accounts; the companion technical note on the Kaplan-Meier estimator derives the continuous-time version from the censored likelihood.

One subtlety decides whether it is estimable at all. If every loan originates in the same year then age and calendar year are the same variable, and no quantity of data can separate seasoning from the economy. Identification requires *staggered origination*, several cohorts observed over overlapping years, so that loans of the same age meet different economies. This is the credit instance of the age-period-cohort problem, and it constrains how the sample must be assembled rather than how the model is fitted.

Simulating six cohorts over ten calendar years, about 75,000 loan-years, recovers the data-generating parameters closely: the fitted hazard is $-3.138 + 0.600a - 0.099a^2 + 0.455m$ against a true $-3.100 + 0.550a - 0.090a^2 + 0.450m$. Evaluating it under three macroeconomic paths gives lifetime ECL of \$125.49 (benign, $m = -1.0$), \$181.79 (base, $m = 0$), and \$292.82 (adverse, $m = +1.5$); weighting these 30/50/20 gives \$187.11.

Two results deserve attention. The weighted figure lands within a few dollars of the flat matrix's \$188.70, so the two approaches agree about the central case, but only the hazard model produces the other two columns, and the \$125-to-\$293 spread is the entire content of a forward-looking provision. Second, expected loss is *convex* in the macroeconomic factor, so the average of the scenario losses exceeds the loss at the average scenario: the weighted \$187.11 exceeds the base-only \$181.79 by 2.9%. A bank modeling only its central forecast under-provides by that margin in calm conditions and by far more when the scenario spread widens, when it matters most. This is Jensen's inequality doing supervisory work, and it is why the standard insists on probability-weighting rather than a single best estimate.

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Two extensions complete the picture. Borrowers prepay, and a loan that prepays generates no credit loss, so lifetime loss is properly a *competing risks* problem against prepayment; ignoring it overstates provisions on the exposures most likely to disappear before defaulting. And the significant-increase test is deliberately left undefined by the standard, making it a threshold-selection problem under the asymmetric costs Chapter 21 works through explicitly.

9.4 Credit Portfolios and Bank Capital

A loan book is not the sum of its loans; this part covers correlated losses, the models that price them, and the regulatory capital held against them.

9.4.1 Portfolio Credit Risk and Economic Capital

Expected loss is what a lender should expect to lose on average, already reflected in loan pricing and loss reserves. It is not by itself a risk measure, since an average says nothing about how much losses could vary in a bad year. Economic capital is the buffer held to absorb unexpected losses beyond the expected level, so that the institution remains solvent in a bad year.

Consider a simplified portfolio of 100 loans, each identical to the Chapter 5 bond example (EAD of \$1,000, PD of 3%, LGD of 60%), with defaults treated, for illustration, as independent across borrowers. The portfolio's expected loss is simply $100 \times \$18 = \$1{,}800$. Treating the number of defaults as a binomial random variable with $n = 100$ and $p = 0.03$, its standard deviation is $\sqrt{np(1-p)} \approx 1.706$ defaults, so the standard deviation of the dollar loss is

$$\hat{\sigma}_{\text{Loss}} = 1.706 \times 0.60 \times 1{,}000 \approx \$1{,}023.52.$$

Approximating the loss distribution as normal (a simplification discussed further below), the 99% Value at Risk of portfolio losses is $\text{EL} + z_{0.99}\,\hat{\sigma}_{\text{Loss}} \approx 1{,}800 + 2.326(1{,}023.52) \approx \$4{,}181$, and economic capital is defined as the unexpected loss beyond the expected level,

$$\text{Economic Capital} = \text{VaR}_{99\%}(\text{Loss}) - \text{EL} \approx 4{,}181 - 1{,}800 \approx \$2{,}381,$$

roughly 1.32 times the portfolio's expected loss. The independent-default assumption is a simplification made for tractability. In reality defaults are correlated: borrowers share the same macroeconomic conditions, and a recession raises every borrower's default probability at once. That makes the loss distribution's tail much fatter than the binomial approximation suggests, which is why regulatory capital models, such as the Basel Committee's asset-correlation framework, build in a correlation parameter. Regulatory capital, the minimum a bank must hold under frameworks such as Basel III, is similar in spirit but set with standardized formulas and correlation assumptions rather than a bank's own estimates, so that banks cannot lower measured risk by choosing favorable assumptions.

Correlation is not the only way a portfolio's risk can exceed what its expected loss suggests; concentration, how evenly exposure is spread across borrowers, matters even before correlation. The Herfindahl-Hirschman Index (HHI), $\text{HHI} = \sum_i s_i^2$ where s_i is borrower i's share of total exposure, quantifies this directly. The 100-loan portfolio above, with exposure spread evenly ($s_i = 1/100$ for every loan), has $\text{HHI} = 100 \times (0.01)^2 = 0.01$. Now compare a differently structured portfolio with the same \$100,000 total exposure and the same 3% average PD, but concentrated in only 10 borrowers of \$10,000 each: $\text{HHI} = 10 \times (0.10)^2 = 0.10$, ten times higher. Both portfolios have the same expected loss of \$1,800, since expected loss depends only on total exposure and average PD, but the concentrated one is riskier. A single default now removes 10% of the exposure rather than 1%, so one or two unlucky defaults among ten large borrowers can produce a loss far outside what the binomial approximation, built for many small exposures, suggests. Regulators and internal risk models therefore impose single-borrower and single-industry concentration limits as a complement to the correlation-adjusted capital of the Vasicek model below: a portfolio can satisfy every correlation assumption in a capital model and still be dangerously undiversified in a way that model, built around a representative borrower, never sees.

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9.4.2 Correlated Defaults: The Single-Factor (Vasicek) Model

The independent-default assumption is the simplification this chapter's case vignette shows going badly wrong at scale. The standard way to relax it, the one underlying the Basel IRB capital formulas, is the single-factor (Vasicek) model. Each borrower's default is driven by an unobserved asset value that depends on a common systematic factor Z, shared by every borrower, and an idiosyncratic factor, with asset correlation ρ controlling how much of each borrower's fortunes are tied to the shared factor. Conditional on the realized systematic factor, defaults become independent, and standard results give the α-quantile of the portfolio's default rate directly as a closed-form worst-case default rate (WCDR),

$$\mathrm{WCDR}(\alpha) = N\left(\frac{N^{-1}(\mathrm{PD}) + \sqrt{\rho}\,N^{-1}(\alpha)}{\sqrt{1-\rho}}\right). \tag{9.9}$$

The formula is quoted rather than proved here. The companion technical note on the single-factor model derives it: conditioning on the systematic factor makes defaults independent, the law of large numbers then collapses an infinitely granular portfolio's loss rate to a strictly monotone function of that factor, and the monotonicity lets the loss distribution be inverted in closed form. The note also verifies that the distribution has mean exactly PD while being strongly right-skewed: correlation moves risk into the tail without changing the average loss. Applying this to the same 100-loan portfolio (PD = 3%) with an illustrative asset correlation of $\rho = 0.20$, typical of the range Basel's own corporate risk-weight formulas use, at the 99% confidence level,

$$\mathrm{WCDR}(0.99) = N\left(\frac{N^{-1}(0.03) + \sqrt{0.20}\,N^{-1}(0.99)}{\sqrt{0.80}}\right) = N\left(\frac{-1.881 + 0.447(2.326)}{0.894}\right) \approx 17.37\%,$$

more than double the 6.97% figure the independent-default binomial approximation implied at the same 99% confidence level. Translating this into dollar terms, the correlated model's worst-case portfolio loss is $100 \times 0.1737 \times 0.60 \times 1{,}000 \approx \$10{,}422$, against the same \$1,800 expected loss as before, giving economic capital of $10{,}422 - 1{,}800 \approx \$8{,}622$, roughly 3.6 times the \$2,381 figure the independent-default assumption produced. This more-than-tripling of required capital is why regulators do not allow banks to assume independent defaults, and it is the mechanism of this chapter's case vignette: a shared systematic factor, whether a national housing downturn or a broader recession, pushes every borrower's default probability the same way at once, fattening the tail far beyond what a model of isolated coin flips suggests, even when every loan's own PD is estimated correctly.

Figure 9.3 shows the whole distribution rather than one quantile of it. With independent defaults, $\rho = 0$, an infinitely granular portfolio would lose exactly 3% every year, and the 100-loan portfolio's binomial distribution stays tightly bunched around that value: the probability that more than 10% of the loans default is 0.021%. At $\rho = 0.20$ the mean is still exactly 3%, but the distribution is pulled apart. Its mode sits near 0.25% and its median at 1.77%, so in most years the portfolio loses *less* than its expected loss, and the probability of a default rate above 10% is 5.02%, one year in twenty. A lender who judges a correlated book by the run of good years it usually produces is looking at the left half of this picture; the capital exists for the right half.

Correlation and stress compound rather than add. Applying Section 9.2.3's stressed PD = 8% to this same portfolio, still holding $\rho = 0.20$, gives

$$\begin{aligned}
\text{Stressed expected loss} &= \$4{,}800,\\
\text{Stressed WCDR}(0.99) &\approx 34.17\%,\\
\text{Worst-case loss} &= 100 \times 0.3417 \times 0.60 \times 1{,}000 \approx \$20{,}504,\\
\text{Economic capital} &= 20{,}504 - 4{,}800 \approx \$15{,}704,
\end{aligned}$$

roughly 6.6 times the base-case, independent-default economic capital of \$2,381. None of the three ingredients behind that 6.6× figure (the stressed PD, the stressed LGD embedded in expected loss, the asset correlation) is individually extreme. A bank that stresses PD and LGD while assuming independent defaults, or that accounts for correlation only under calm-period PD, captures a fraction of the combined effect, which is why regulators require the factors to be modeled jointly.

The single-factor model carries one further simplification: the closed-form WCDR assumes an infinitely granular portfolio of many small, roughly equal exposures, so that no single default materially moves the

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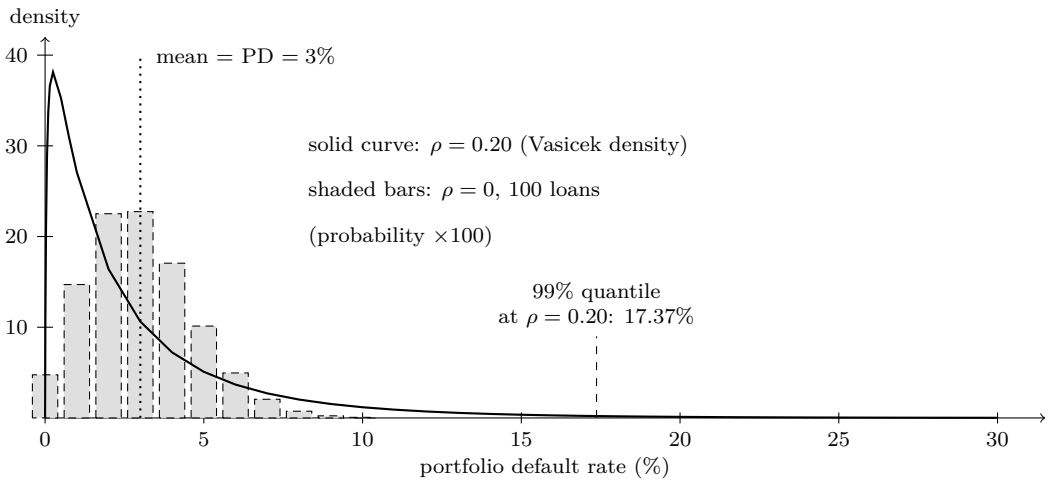


Figure 9.3: The distribution of the portfolio default rate at PD = 3%: the Vasicek density at asset correlation $\rho = 0.20$ against the 100-loan portfolio's binomial distribution under independence ($\rho = 0$), both with mean 3%

total loss. A real loan book is never quite that diversified. A handful of unusually large exposures adds name concentration risk the formula cannot see: if one of the 100 loans were ten times its size, its default would move portfolio losses far more than the granular formula assumes, even with every PD, LGD, and correlation fixed. Basel addresses this with a separate granularity adjustment, an add-on charge that grows as exposures concentrate among a few large borrowers. The closed-form result is only as trustworthy as its granularity assumption, one more instance of the chapter's theme: every formula is exact given its assumptions, and credit risk management lies largely in knowing which assumption is closest to breaking for a real portfolio.

9.4.3 The Basel Framework: A Brief History

Today's regulatory capital requirements are the product of a decades-long response to successive financial crises, summarized in Table 9.6.

The finalization's national rollout has been slower and more uneven than the Basel calendar suggests, and the dates are still moving. As of October 2026, the European Union has applied the new credit and operational-risk rules since 1 January 2025 through its Capital Requirements Regulation (CRR3), with the output floor starting at 50% and rising to 72.5% in 2030 and some transitional relief running to 2032 (European Parliament and Council of the European Union 2024). The United Kingdom's Prudential Regulation Authority finalized its version, "Basel 3.1," in January 2026, to take effect on 1 January 2027 with the floor reaching 72.5% in 2030 (PRA 2026). In the United States, the agencies withdrew their 2023 "Basel III endgame" proposal and issued a re-proposal on 19 March 2026, whose comment period closed on 18 June 2026; no final rule had been adopted as of October 2026 (Board of Governors of the Federal Reserve System et al. 2026). Chapter 8 tracks the corresponding, and even more delayed, market-risk timetable.

Each accord reacted to its predecessor's weaknesses. Basel I's uniform risk weights did not distinguish a safer borrower from a riskier one, which Basel II addressed by permitting internal models. The 2008 crisis then revealed that those models, calibrated on the calm pre-crisis period, had let some banks hold far too little capital against risks that materialized suddenly and severely, the model-risk and procyclicality concerns of this chapter's challenges section, which Basel III's higher and higher-quality capital requirements were designed to address. Basel II's internal-ratings-based approach is built on this chapter's machinery. A bank estimates its own PD (Section 9.2.6), LGD (Section 9.2.2), and EAD for each exposure and feeds them into a regulatory risk-weight formula derived from the Vasicek model of Section 9.4.2, with the asset correlation set by the regulator rather than estimated by the bank, so that a bank cannot lower its capital by assuming a lower correlation than its portfolio exhibits.

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Accord	Key development
Basel I (1988)	Introduced a simple, standardized minimum capital ratio (8% of risk-weighted assets) applied uniformly across large internationally active banks
Basel II (2004)	Allowed large banks to use internal PD, LGD, and EAD models (much like Sections 9.2.1 and 9.2.6 of this chapter) to compute risk-weighted assets, rather than the fixed weights of Basel I
Basel III (2010–2017, phased in)	Raised capital quality and quantity requirements substantially in response to the 2008 financial crisis, added new liquidity requirements (Chapter 8's LCR and NSFR), and introduced the shift from VaR to Expected Shortfall for market-risk capital discussed in Chapter 8
Basel III finalization (2017; Basel start date 2023, national dates later)	Limited how far internal models can lower capital, through an output floor phasing in to 72.5% of standardized risk-weighted assets (by 2028 on the Basel schedule; national timetables in the text below), and replaced all of Basel II's operational-risk approaches with a single Standardized Measurement Approach

Table 9.6: A brief history of the Basel bank capital accords

9.4.4 Operational Risk Capital: The Basic Indicator Approach

Operational risk, introduced from the market-risk side in Chapter 8, is harder to quantify than market or credit risk, since no return series or default history applies uniformly across firms. A rogue trader's unauthorized position, a failed core banking system upgrade, and a mis-sold financial product are all operational risk events, but they share little statistical structure.

Basel II's simplest treatment, the Basic Indicator Approach, now of historical interest but still the clearest illustration, sidestepped this difficulty by tying capital to firm size rather than to any operational loss model,

$$K = \alpha \times \overline{\mathrm{GI}},$$

where $\overline{\mathrm{GI}}$ is the bank's average annual gross income over the past three years (a simple proxy for the overall scale of its operations) and $\alpha = 15\%$ is a fixed regulatory parameter. Under that approach, a bank with average annual gross income of \$200 million would hold

$$K = 0.15 \times 200{,}000{,}000 = \$30{,}000{,}000$$

in operational risk capital, however many operational loss events it has experienced.

Basel II also offered a finer standardized variant and an internally modeled Advanced Measurement Approach, but the 2017 Basel III finalization retired all three in favor of a single Standardized Measurement Approach, which scales capital with a size-based business indicator adjusted by the bank's own loss history; no bank may now set operational-risk capital with its own internal model, a reversal Chapter 10 examines. The size-based proxy illustrates a broader principle: when a risk is too hard to model credibly and consistently across institutions, regulators fall back on a simple, conservative, hard-to-manipulate formula rather than a model whose inputs a bank could shade in its favor, the tension between sophistication and gameability that runs through the internal-ratings-based approach to credit risk. The contrast with the rest of this chapter is instructive. PD, LGD, and asset correlation are hard to estimate, but a bank has a coherent statistical framework (logistic regression, the Merton model, the Vasicek model) for estimating each from data that plausibly exists. Operational risk events, a rogue trader, a failed software deployment, a mis-sold product, share so little common structure that no single statistical framework fits all of them well, which is why regulators settled for a size-based proxy rather than an operational-risk analogue of the PD/LGD/EAD decomposition. Chapter 10 develops both the loss-distribution machinery behind that capital charge and the AI methods (incident-text classification, anomaly detection, surveillance, early warning) a bank uses to manage it.

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9.5 Counterparty Credit Risk and Valuation Adjustments

A loan's exposure is known; a derivative's is not. This part prices the credit risk of a contract whose value to the bank, and therefore what the bank stands to lose, moves with the market.

9.5.1 Exposure Profiles: Expected and Potential Future Exposure

Everything so far has treated EAD as a number fixed, or at worst drawn down, in advance. An over-the-counter derivative breaks that assumption. An interest-rate swap is worth nothing to either side on the day it is struck, and thereafter its value drifts with interest rates, sometimes in the bank's favor and sometimes against it. If the counterparty defaults while the swap is worth $V > 0$ to the bank, the bank loses (a fraction LGD of) V, since it must replace the contract in the market at that price; if the counterparty defaults while $V < 0$, the bank still owes the full amount to the defaulted counterparty's estate and gains nothing. The exposure is therefore $\max(V, 0)$, a call-option payoff on the contract's own future value, and it is random. This is counterparty credit risk, and it is the credit risk of every derivatives desk in the book.

Two summaries of that random exposure do most of the work. The *expected exposure* at a future date t is the mean of the positive part,

$$\mathrm{EE}(t) = \mathbb{E}[\max(V(t), 0)]\,, \tag{9.10}$$

which is the input to pricing, while the *potential future exposure* $\mathrm{PFE}_\alpha(t)$ is a high quantile of the same distribution, typically $\alpha = 95\%$ or 97.5%, which is the input to credit limits, the counterparty analogue of a Value at Risk. Plotted against t, either one is an *exposure profile*, and its shape is determined by the instrument. For a swap, two forces pull in opposite directions: the uncertainty in rates grows with time, roughly as $\sqrt{t}$, while the number of remaining payments that a rate move can affect shrinks to zero at maturity. The product of the two gives the characteristic hump.

That hump can be written down in closed form under one simplifying assumption. Let rates move as an arithmetic Brownian motion with a normal volatility of σ per year (the Bachelier, or normal, model of rates), and approximate the value of a payer swap with notional N and maturity T, struck at the market rate, by the rate move times the remaining annuity, ignoring discounting within it. Then $V(t)$ is normal with mean zero and standard deviation $s(t)$, and the standard normal integral $\mathbb{E}[\max(Z, 0)] = \varphi(0) = 1/\sqrt{2\pi} \approx 0.39894$ gives

$$s(t) = N\sigma\sqrt{t}\,(T - t), \qquad \mathrm{EE}(t) = 0.39894\, s(t), \qquad \mathrm{PFE}_{95\%}(t) = 1.645\, s(t). \tag{9.11}$$

The function $\sqrt{t}\,(T - t)$ peaks at $t = T/3$, so a swap's exposure is largest about a third of the way through its life. For a five-year, \$10 million swap with $\sigma = 1\%$ (100 basis points of normal volatility per year), evaluated at the midpoint of each year, $t = 0.5, 1.5, \ldots, 4.5$, the standard deviations $s(t)$ are \$318,198, \$428,661, \$395,285, \$280,624, and \$106,066, and the expected exposures follow in Table 9.7. At $t = 1.5$, for example, $s = 10{,}000{,}000 \times 0.01 \times \sqrt{1.5} \times 3.5 = \$428{,}661$, so $\mathrm{EE} = 0.39894 \times 428{,}661 = \$171{,}011$ and $\mathrm{PFE}_{95\%} = 1.6449 \times 428{,}661 = \$705{,}084$, more than four times the expected exposure. Figure 9.4 draws both profiles. A production system, facing hundreds of trades of different types with one counterparty, cannot use a closed form, so it simulates thousands of paths of every market risk factor, revalues every trade on every path at every future date, and reads the profiles off the simulated distribution. The companion notebook does this for the one swap, with 200,000 seeded paths, and recovers the closed-form expected exposures to within about 0.6%.

9.5.2 The Credit Valuation Adjustment

A derivative's textbook price, the one Chapter 5 computes, assumes that both parties perform. The *credit valuation adjustment* (CVA) is the amount by which the value of the contract must be reduced because the counterparty might not: the market price of the counterparty credit risk. It is the expected-loss formula of Equation (9.1) applied period by period, with the expected exposure in place of a fixed EAD, a marginal default probability for each period in place of a single PD, and discounting, since the loss falls in the future. Dividing the life of the trade at dates $t_0 = 0 < t_1 < \cdots < t_n = T$,

$$\mathrm{CVA} \approx \mathrm{LGD} \sum_{i=1}^{n} \mathrm{EE}(t_i)\, \mathrm{PD}(t_{i-1}, t_i)\, \mathrm{DF}(t_i), \tag{9.12}$$

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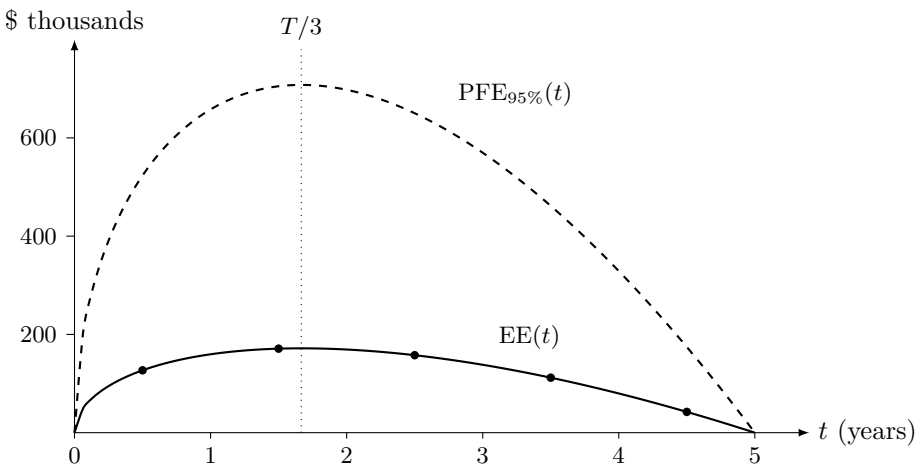


Figure 9.4: Exposure profiles of a five-year, $10 million payer swap under 1% normal rate volatility: expected exposure (solid, with the five dates used for CVA marked) and 95% potential future exposure (dashed). Both peak at a third of the swap's life.

where $\mathrm{PD}(t_{i-1}, t_i)$ is the probability that the counterparty defaults in the ith period and $\mathrm{DF}(t_i)$ is the discount factor. With a constant hazard rate h, the CDS-implied quantity of Section 9.3.3, the marginal probabilities are differences of survival probabilities, $\mathrm{PD}(t_{i-1}, t_i) = e^{-ht_{i-1}} - e^{-ht_i}$. The formula treats exposure and default as independent, which allows the expectation of their product to be written as a product of expectations; Section 9.5.5 examines what happens when they are not.

Equation (9.12) also explains why CVA is priced from *market-implied*, not historical, default probabilities. A bank that wants to hedge its CVA buys CDS protection on the counterparty, so the relevant probability is the one at which that protection trades. CVA therefore moves every day with the counterparty's credit spread even when nobody defaults, and that is where most of the losses fell in the 2008 crisis: by the Basel Committee's estimate, roughly two-thirds of banks' losses attributed to counterparty credit risk were mark-to-market CVA losses and only about one-third were actual defaults, which is why Basel III attached a separate capital charge to CVA volatility (BCBS 2011).

9.5.3 Netting, Collateral, and the Margin Period of Risk

Two contractual devices cut counterparty exposure, often sharply. The first is *netting*: under a master agreement, all the trades with one counterparty are closed out together in default and only the net amount is owed, so the exposure is $\max(\sum_j V_j, 0)$ rather than $\sum_j \max(V_j, 0)$, and trades that move in opposite directions offset one another. For two trades whose values are normal with the same standard deviation and correlation ρ, the netted expected exposure is $\sqrt{(2+2\rho)}/2$ times the sum of the two standalone expected exposures: 0.707 for independent trades, 0.500 for trades with $\rho = -0.5$, and no benefit at all only when $\rho = 1$. This is why exposure is always computed per *netting set*, never trade by trade.

The second device is *collateral*. Under a variation-margin agreement, the party that is out of the money posts cash or securities equal to the contract's value, usually daily, so a default leaves the surviving party holding collateral against its claim. Agreements often allow a *threshold* H, an amount of uncollateralized exposure each side tolerates before calling for margin, in which case the exposure becomes $\min(\max(V, 0), H)$: everything above the threshold is covered. For a normally distributed value with standard deviation s, the expected collateralized exposure has a closed form,

$$\mathbb{E}[\min(\max(V, 0), H)] = s\,[\varphi(0) - \varphi(H/s)] + H\,[1 - N(H/s)]\,, \tag{9.13}$$

which falls to zero as $H \to 0$ and recovers $s\,\varphi(0)$ as $H \to \infty$.

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Even a zero threshold does not remove the exposure entirely, because collateral arrives with a lag. Between the last margin call a defaulting counterparty met and the moment the surviving bank has closed out and replaced the trades, the *margin period of risk*, conventionally ten business days for bilateral trades, the market keeps moving and the collateral stops being topped up. The residual exposure is the value change over that window, whose standard deviation is $s(t)$ scaled by $\sqrt{\text{MPoR}/t}$ rather than the full $s(t)$, which is far smaller but never zero.

9.5.4 Worked Example 9.2: The CVA of a Five-Year Interest-Rate Swap

Price the counterparty risk on the swap of Section 9.5.1, traded with a counterparty whose one-year CDS trades at the 200 basis points of Section 9.3.3, so that $h = 0.02/(1 - 0.40) = 3.333\%$, with the chapter's LGD $= 60\%$ and the 5% discount rate of Table 9.5. Each year is one default bucket, with exposure evaluated at its midpoint. Year one's marginal default probability is $1 - e^{-0.03333} = 3.2784\%$ and its discount factor is $1.05^{-0.5} = 0.9759$, so its contribution is

$$0.60 \times 126{,}943 \times 0.032784 \times 0.9759 = \$2{,}436.83,$$

and Table 9.7 repeats the calculation for the other four years.

Year	t_i	EE(t_i)	PD(t_{i-1}, t_i)	DF(t_i)	LGD × EE × PD × DF
1	0.5	\$126,943	3.2784%	0.9759	\$2,436.83
2	1.5	\$171,011	3.1709%	0.9294	\$3,023.95
3	2.5	\$157,696	3.0670%	0.8852	\$2,568.65
4	3.5	\$111,953	2.9664%	0.8430	\$1,679.79
5	4.5	\$42,314	2.8692%	0.8029	\$584.84
				CVA	\$10,294

Table 9.7: The credit valuation adjustment of a five-year, \$10 million payer swap, uncollateralized

The CVA is \$10,294, about 10.3 basis points of notional paid up front, or 2.38 basis points a year spread over the swap's life, which is the amount a dealer adds to the rate it quotes this counterparty. The profile of the contributions follows the exposure hump, discounted and thinned by survival, with year two the largest. The Monte Carlo exposures of the companion notebook give \$10,258, and a common back-of-the-envelope shortcut, the CDS spread times the discounted expected exposures, $0.02 \times \sum_i \text{EE}(t_i)\,\text{DF}(t_i) = \$11{,}015$, lands within 7%, since it is the credit triangle of Equation (9.7) again: $s \approx h(1 - R)$.

Now suppose the two parties sign a variation-margin agreement with a threshold of $H = \$250{,}000$. At $t = 1.5$, $H/s = 250{,}000/428{,}661 = 0.583$, and Equation (9.13) gives an expected exposure of \$96,714, down from \$171,011. Repeating this at every date gives expected exposures of \$87,718, \$96,714, \$94,472, \$83,295, and \$41,986, and a CVA of \$6,763, 65.7% of the uncollateralized figure. The threshold helps least where exposure is already small: in year five the swap's value rarely exceeds \$250,000, so collateral is almost never called and the expected exposure barely changes. Removing the threshold altogether, so that only the ten-day margin period of risk remains, cuts the year-one expected exposure to \$35,905 and the CVA to \$1,743, 16.9% of the uncollateralized value. The two figures bracket the practical range: collateral removes most counterparty risk, a threshold gives part of that back, and no agreement removes the gap risk of the closeout window.

9.5.5 Wrong-Way Risk as a Correlation Stress

The independence assumption in Equation (9.12) fails in the cases that matter most. *Wrong-way risk* is a positive dependence between exposure and the counterparty's default probability: the trade is worth most to the bank in the states in which the counterparty is most likely to fail. A bank that bought CDS protection on one bank from another whose fortunes are tied to the first holds a contract worth most when its seller is weakest. A bank that receives dollars and pays local currency in a cross-currency swap with an

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emerging-market bank finds the swap most valuable after a collapse of the local currency, the same event most likely to push its counterparty into default. Its opposite, *right-way risk*, arises when exposure rises as default risk falls.

The single-factor model of Section 9.4.2 gives a clean way to quantify it, as a correlation stress. Model the counterparty's default in period i as its standardized latent variable X falling below $k_i = N^{-1}(\mathrm{PD}(t_{i-1}, t_i))$, as in Equation (9.9), and let the swap's standardized value $Y = V/s$ have correlation c with X, with $c < 0$ meaning wrong-way risk. Defaults happen in the lower tail of X, so they now come with a shifted exposure distribution. In closed form, the conditional mean of Y given default is $\mathbb{E}[Y \mid X < k] = -c\,\varphi(k)/N(k)$. For year one, $k_1 = N^{-1}(0.032784) = -1.841$ and $\varphi(k_1)/N(k_1) = 2.233$, so with $c = -0.3$ the swap's value at the moment of default is centered 0.670 standard deviations above zero rather than at zero. The expected exposure *given* default, computed in the companion notebook by integrating over the conditional distribution, is then 2.02 times the unconditional figure, and applying the corresponding multiplier in every year doubles the CVA, to \$20,952. A correlation of only -0.1 already raises it to \$13,417, and -0.5 to \$30,154.

The multiplier is large because default is a tail event, and conditioning on a tail event drags any correlated variable a long way. This is the Gaussian copula lesson of this chapter's case vignette in a different instrument: a correlation that looks modest moves the loss distribution's tail by a large factor, and it cannot be estimated with any confidence from data in which the counterparty never defaults. Regulators answer the same way they answer the asset-correlation problem, with a conservative supervisory parameter rather than a bank's own estimate: *specific* wrong-way risk, where the dependence is structural, such as a counterparty selling protection on itself or an affiliate, must be identified and treated separately, and the general multiplier described next is set conservatively partly with this effect in mind.

9.5.6 From CVA to XVA: Capital, and Where Adjoint Differentiation Pays

Regulation prices counterparty risk twice. The exposure measure for default capital is the Basel Committee's standardized approach for counterparty credit risk (SA-CCR), which sets EAD $= 1.4 \times$ (replacement cost + PFE add-on) with supervisory formulas for netting and collateral (BCBS 2014), and the separate charge for CVA volatility, rebuilt in the 2017 finalization as a basic approach and a sensitivity-based standardized approach and recalibrated in July 2020 (BCBS 2020), is part of the Basel III finalization whose national timetables Section 9.4.3 records. Banks also now price more than credit. Alongside CVA sit a debit valuation adjustment for the bank's own default risk, a funding valuation adjustment for the cost of funding uncollateralized positions, a margin valuation adjustment for the cost of posting initial margin, and a capital valuation adjustment for the cost of the capital the trade consumes, collectively XVA (Gregory 2020). Every one of them requires the same simulation: market paths, a revaluation of every trade on every path at every date, and an aggregation by netting set.

That is where the computational tools of Chapter 8 earn most of their keep. A CVA desk hedges its CVA, and the standardized CVA capital charge is computed from CVA sensitivities, so the bank needs the derivative of a whole-portfolio, simulation-based number with respect to every curve point and volatility, often thousands of inputs. Bumping and rerunning the simulation once per input is infeasible; adjoint algorithmic differentiation delivers all of them for a small multiple of one run, which is why XVA systems were among its earliest large-scale adopters. The inner revaluations are the other bottleneck, since a path-dependent or callable trade cannot be priced in closed form at each simulated node, and the standard remedy is to replace the inner pricer with a fitted approximation, least-squares regressions on the simulated state or neural-network surrogates, the same move Chapter 11 describes for insurers' nested valuations. Huge and Savine (2020) joined the two ideas in *differential machine learning*: since adjoint differentiation produces each simulated path's pathwise derivatives almost for free, a network can be trained on both the simulated values and their derivatives, so that it learns the pricing function and its Greeks together, from far fewer simulations than training on values alone requires. It is the clearest case in this book of a machine-learning technique valuable because a financial model already supplies the gradients that other fields would have to estimate.

9.6 Machine Learning and Model Governance in Credit Risk

Every technique so far (logistic regression, the Merton model, transition matrices, the Vasicek model) is a classical statistical or structural model, chosen because a credit decision affecting a real borrower must be

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explainable to that borrower, to a regulator, and to a model validator. Machine learning nonetheless enters credit modeling in several concrete ways that extend rather than replace these tools.

Tree ensembles and gradient boosting, introduced in Chapter 6, routinely outperform logistic regression on raw discriminatory power, capturing nonlinear interactions (a debt-to-income ratio that matters far more once credit history falls below some threshold, say) that a linear logistic regression cannot represent. Chapter 21's capstone works through this comparison on a real consumer credit dataset, finding a several-point ROC-AUC improvement, at the cost of the closed-form Shapley decomposition Chapter 18 showed is available only for a model linear in the log-odds.

9.6.1 A Head-to-Head Test: Gradient Boosting versus Logistic Regression

Section 9.2.6's six-applicant sample is too small, and too cleanly separated, to show a tree-ensemble advantage; logistic regression already reaches a perfect in-sample AUC of 1.0 on it. The companion notebook instead simulates a population of 10,000 borrowers with a threshold interaction built into the true default process: debt-to-income drives default risk, but far more steeply once credit history falls below four years, a nonlinearity a logistic regression linear in both features cannot represent.

Splitting into 8,000 training and 2,000 held-out borrowers, a logistic regression fit on debt-to-income and credit history alone achieves a held-out AUC of 0.8494; a gradient-boosted tree ensemble fit on the identical two features, with no additional information, reaches a held-out AUC of 0.8634, an improvement of 1.40 percentage points, consistent in direction (if more modest in this smaller, controlled setting) with the several-point improvement of Chapter 21's real-dataset capstone. The gradient-boosting model's feature importances show why: it assigns 83% of its importance to credit history and only 17% to debt-to-income, because credit history does double duty in the true model, as a direct risk factor and as the threshold variable that determines how much debt-to-income matters, a role a single linear coefficient cannot capture.

This result sits alongside Section 9.2.9's, where a scorecard beat a gradient-boosted model, and reconciling the two is the whole argument for binning. The logistic regression beaten here is *linear in both features*, so a threshold effect is what it cannot represent. The scorecard that wins there is not: weight-of-evidence binning gives each band its own value, so the threshold is absorbed. What the tree ensemble buys here is flexibility over a straight line, not over a scorecard, and the gap closes once the straight line is replaced by the form lenders deploy. A tree's residual advantage after binning is whatever interactions remain, which in credit data are usually modest. As Section 9.2.7 emphasized, the improvement is meaningful only because it is measured on held-out data; on the 8,000 training borrowers, the ensemble's flexibility alone would inflate the gap whether or not the interaction were real.

Alternative data sources (satellite imagery, transaction and cash-flow data, web activity), developed in Chapter 17, extend credit scoring to borrowers with too little credit-bureau history for conventional scoring, a distinct capability rather than a more flexible functional form. Cash-flow underwriting makes the distinction concrete: its raw material is a stream of semi-structured merchant strings, so there are no bins to choose until a text classifier of the kind Chapter 16 develops has decided which transactions are rent, and the machine learning sits at the front of the pipeline rather than at the modeling step. Graph neural networks, introduced in Chapter 17 for a shell-company layering network, apply naturally to counterparty and supply-chain credit risk, since a firm's default risk depends partly on the health of its major customers and suppliers, a network effect no borrower-level feature table can represent. In every one of these applications the discipline of Section 9.2.7 applies: discrimination and calibration must both be validated out of sample, and added flexibility is worth its added opacity only if validation and monitoring can keep pace.

9.6.2 Monotonic Constraints: Making a Tree Ensemble Defensible

When a tree ensemble is the right choice, it still needs the property Section 9.2.8's scorecard gets for free. A boosted tree fitted to credit data will somewhere do something indefensible: a finite sample always contains regions where noise makes a few high-leverage borrowers look safer than those just below them, and the model learns the wiggle. If predicted default probability falls as debt-to-income rises, then somewhere in the book is an applicant who would be approved by borrowing *more*, and a declined applicant whose adverse action notice cannot truthfully be written.

Monotonic constraints tell each tree that splits on a feature may move the prediction in only one direction, so the fitted function is monotone by construction rather than by inspection. Refitting the simulation above

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with the constraint, the unconstrained model reverses direction at 53 of 199 grid steps, and at its worst its fitted probability *falls* by more than twenty percentage points, from 62.0% to 39.6%, across a single grid step in which debt-to-income rises from 46.8% to 47.1%; the constrained model reverses nowhere. The expected accuracy cost does not appear: ROC-AUC is 0.8757 constrained against 0.8597 unconstrained, both below the 0.8812 ceiling. The constraint *improved* discrimination, because the monotone relationship is really present and forbidding reversals removes only overfitting. That is not a general law: a constraint encoding something true costs nothing and regularizes, while one encoding something false costs accuracy and deserves questioning rather than silent removal.

A monotonic constraint is, more generally, an *inductive bias*, prior knowledge about the shape of the solution built into the model class rather than learned from data, in the same family as the translation equivariance a convolutional network gets from weight sharing in Chapter 17.

9.6.3 Fairness in Lending: Searching for a Less Discriminatory Alternative

Fairness in lending is a stronger obligation than the auditing Chapter 18 develops, and one where machine learning is the tool rather than the problem. Under disparate-impact doctrine, showing that a disparity is justified by business necessity does not end the analysis: if a *less discriminatory alternative* exists, a model serving the lender's purpose comparably well with a smaller disparity, then using the original is not defensible. In U.S. consumer credit the federal footing for this doctrine narrowed in 2026, when the CFPB amended Regulation B to remove the disparate-impact "effects test" under the Equal Credit Opportunity Act, effective 21 July 2026 (CFPB 2026); disparate treatment remains prohibited, and state enforcers, the Fair Housing Act, and regimes abroad still reach disparate outcomes (Massachusetts Attorney General 2025), so the search described here remains sound practice (Chapter 18). This converts fairness from a test into a *search* over the model space, conducted by retraining under fairness regularization, transforming features that carry disparity without much signal, and varying thresholds and segmentations to trace the achievable frontier of performance against disparity. It is the third item in Section 9.2.9's corollary, and the one setting in credit where fitting many models rather than one is the compliant behavior rather than the suspicious one. The European Union now writes a comparable obligation into statute: its Artificial Intelligence Act classifies AI systems used to evaluate the creditworthiness of natural persons or to establish their credit score (other than for detecting financial fraud) as high-risk, subjecting them to risk-management, data-governance, documentation, and human-oversight requirements (EU 2024) that the Digital Omnibus has deferred to 2 December 2027 (EU 2026); Chapter 18 sets out that regulatory landscape in full.

9.6.4 Model Validation and Governance for Credit Models

Chapter 8 developed the general model risk management framework (independent validation, ongoing monitoring, model inventory and documentation), and its three-lines-of-defense structure applies unchanged to credit models: a lending or trading desk originates the risk (first line), an independent credit risk function builds and monitors the PD, LGD, and rating models (second line), and internal audit periodically verifies that the process works as designed (third line).

Credit models carry two validation concerns beyond the general framework. First, discriminatory power (Section 9.2.7's AUC and Gini coefficient) must be checked out of sample, not on the training data. Second, calibration, whether predicted probabilities match observed default rates, is distinct from discrimination, and a high AUC does not guarantee it: a model can rank borrowers correctly while being systematically over- or under-confident about the probability level, the calibration warning of Chapter 6. A validator must check both, since a model that passes discrimination but fails calibration can still produce a badly wrong dollar reserve.

A mature validation function also revisits, on a regular cycle, the chapter's other structural assumptions. Rating transition matrices, Section 9.4.1's independent-default assumption, and Section 9.4.2's asset correlation are all estimated quantities that can drift silently as conditions change: a transition matrix estimated from a benign decade understates downgrade risk once conditions deteriorate, and an asset correlation that looked reasonable in a calm period is the kind of assumption this chapter's case vignette showed failing at the worst moment. Validation is therefore an ongoing discipline applied to every input a model depends on, not a one-time check of its headline PD or LGD.

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9.7 Case Vignette: The Gaussian Copula and the Mismodeling of CDO Credit Risk

Chapter 8's case vignette described how correlations that looked stable in calm markets rose sharply during the 2008 crisis. The credit-risk analogue concerns collateralized debt obligations (CDOs), securities that pool many loans, mortgages, or bonds and slice the cash flows into tranches of different seniority, and the correlation model Wall Street and the rating agencies used to price and rate them.

In 2000 the quantitative analyst David X. Li published an influential paper proposing the Gaussian copula as a tractable way to model default correlation across many borrowers. A single correlation parameter summarized how likely two borrowers were to default together, turning an intractable joint-default problem into one easily calibrated number, and the approach was quickly adopted to price and rate mortgage-backed CDO tranches.

The difficulty was the one this chapter's economic-capital example flags. The correlation parameter was typically calibrated on a period of calm, rising home prices, and it assumed that mortgage defaults across geographically diverse pools would remain only modestly correlated, largely independent local events. When the nationwide housing downturn arrived, defaults across the country rose together far more than assumed, the "correlations go to one in a crisis" pattern, and AAA-rated CDO tranches, built on the premise that a nationwide wave of correlated defaults was nearly impossible, suffered mass downgrades and defaults in 2007 and 2008. A widely read 2009 press account called the Gaussian copula "the formula that killed Wall Street," not because the mathematics was wrong (a copula is a legitimate way to model dependence) but because its single correlation number was calibrated on calm pre-crisis data that understates tail dependence.

The lesson is easy to oversimplify into "correlation models are untrustworthy." The Gaussian copula and Section 9.4.2's Vasicek model are close relatives: both reduce a joint-default problem to a single correlation acting through a shared systematic factor, and neither is mathematically inconsistent.

What failed was an estimation choice, made once on a sample that never contained a nationwide housing downturn and then treated as fixed rather than stress-tested the way Section 9.2.3 stress-tests PD and LGD. Stressing the correlation toward a more severe scenario, as Section 9.4.2 does for this chapter's loan portfolio, would have revealed a far larger tail loss before 2007, with the same model. The failure was organizational as much as mathematical: nothing in the copula framework prevented stress-testing the correlation, but originators, rating agencies, and investors all benefited in the short run from a favorable, low-correlation rating, so nobody was rewarded for asking the question.

As in market risk, a correlation calibrated on calm historical data is a recurring point of failure, and mathematical sophistication is no substitute for stress-testing the assumption, here default correlation, that matters most when conditions are worst.

9.8 Challenges in Credit Risk Modeling

Several challenges recur across nearly every credit risk application, extending the market-risk challenges of Chapter 8.

Correlated defaults and tail dependence. The independent-default simplification of Section 9.4.1 understates portfolio risk for pedagogical clarity; a real portfolio model must estimate default correlation, the difficulty this chapter's case vignette showed going wrong at scale. Section 9.4.2's single-factor model brings correlation back in but assumes one shared factor; a portfolio spanning several industries and geographies is better described by several correlated factors, and misspecifying how many exist, or how strongly a borrower loads on each, is statistically difficult and highly consequential for capital.

Procyclicality. PD and LGD calibrated on calm data understate risk in good times and spike once a downturn is underway, an amplifying loop, since understated capital encourages more lending when underwriting standards are already loosening. Section 9.2.3's stress test counteracts this by testing the scenario before it arrives. Section 9.4.2's combined calculation, a 6.6× increase in economic capital from three modest-looking assumptions, shows how severely procyclicality compounds when every input moves the same way.

Data quality and limited loss history. Credit data is plentiful in rows and thin in the dimension that constrains a model. A lender may hold records on a million loans, yet the number of complete economic cycles those loans have lived through is one, or zero, and default is a slow event: an origination cohort's true

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loss rate is not known until years after the underwriting decision it would judge. A decade of records can therefore amount to a single draw from the distribution a PD model is meant to describe, which is why a PD estimated from a decade that happened not to include a recession will systematically understate true credit risk. The sample is also selected rather than merely observed, since outcomes exist only for the applicants the lender chose to approve. That is Section 9.2.6's reject-inference problem as a data limitation, and no number of additional rows fixes it, because the missing borrowers are missing by construction. Section 9.2.7's perfect in-sample AUC is the same danger in miniature: with six training observations, apparent discriminatory power says almost nothing about performance on new applicants.

Regulatory and business tension. What makes this tension specific to credit is timing. A market-risk limit breach announces itself within days, whereas a loosened underwriting standard books revenue immediately and delivers its losses two or three years later, so for several reporting periods a credit function being overruled looks like one that was merely too cautious. That is why lending standards reliably loosen late in an expansion, when competition for volume is fiercest and the last downturn furthest away, and it is what the three-lines-of-defense structure of Chapter 8 is meant to counteract, by separating the unit that originates loans from the one that validates the models pricing and reserving them. A second constraint has no market-risk analogue: a consumer credit model must be explainable to the applicant it declines, under the adverse-action rules of Chapter 18, so pressure toward raw accuracy and toward transparency can pull the model's functional form in opposite directions, the trade-off monotonic constraints exist to negotiate.

Regulatory arbitrage. Whenever capital is calculated by a specific formula, whether the Basic Indicator Approach's gross-income proxy or an internal-ratings-based model, an institution has an incentive to minimize the capital charge rather than the underlying risk, a gap between measured and true risk that can widen quietly even when every calculation is technically correct. An institution allowed to choose its own asset correlation in Section 9.4.2's framework, for instance, has an obvious incentive to pick a lower ρ than its portfolio exhibits, understating capital while remaining compliant on paper.

Machine learning adds a further layer of model risk. Tree ensembles and neural networks can capture nonlinear default and loss patterns that logistic regression and the Merton model, with their fixed functional forms, cannot, but they inherit the overfitting and interpretability concerns of Chapter 6 in a setting where undetected errors are unusually costly. An opaque credit model that a regulator or validator cannot interrogate is hard to trust with a bank's capital, whatever its backtested accuracy, so Section 9.2.7's discrimination and calibration checks apply to a gradient-boosted model as much as to a hand-built logistic regression.

9.9 Key Takeaways

Credit risk decomposes into probability of default, loss given default, and exposure at default, whose product gives expected loss. Logistic regression is the standard tool for estimating PD from borrower characteristics, evaluated with the Gini coefficient and AUC for ranking power and separately for calibration. The Merton model and CDS-implied hazard rates estimate PD from market prices (equity or CDS spreads) without a default history.

LGD is not a fixed constant either: the collateral-coverage regression of Section 9.2.2 estimates recovery from a lender's own default history.

Rating transition matrices extend one-period default probabilities to multi-year cumulative estimates, and credit stress testing shows what happens to expected loss when PD and LGD move toward recession values together. That same matrix supplies the PD term structure IFRS 9 and CECL require, turning a one-year expected loss into a lifetime one: for the chapter's Medium-rated exposure the provision rises from \$57.14 to \$188.70, a 3.30× cliff crossed by reclassification alone, and a fitted discrete-time hazard model conditions that figure on the economy, where the convexity of loss in the macroeconomic factor is why the standard demands probability-weighted scenarios rather than a single central forecast.

The chapter's most useful negative result concerns machine learning itself. A credit scorecard, weight-of-evidence bins fed to a logistic regression, is a generalized additive model fitted by hand, the object an Explainable Boosting Machine fits automatically, and on simulated data containing an interaction it lands within 0.003 AUC of the theoretical ceiling with nothing beating it. Machine learning earns its keep in credit by doing what a scorecard cannot: building features from data with no natural bins, producing a term structure rather than a probability, and searching a model space rather than fitting one model. Where a tree

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ensemble is the right tool, monotonic constraints buy back the scorecard's defensibility, here at no cost in accuracy.

Economic capital turns the gap between expected and unexpected loss into a capital buffer, and the Vasicek model of Section 9.4.2 shows how much larger that buffer must be once defaults are correlated. Stress and correlation combined multiplied required capital nearly sevenfold in the worked example, and the case vignette showed the same effect, neglected, contributing to a systemic mispricing of mortgage-backed credit risk.

Two extensions push the same decomposition into territory where its inputs are no longer fixed numbers. A derivative's exposure is a random variable, summarized by expected and potential future exposure profiles, and the credit valuation adjustment is expected loss applied period by period to that profile: \$10,294 on the chapter's \$10 million swap, cut to \$6,763 by a collateral threshold and to \$1,743 by full variation margin, and doubled by a wrong-way correlation of only -0.3. Climate risk, in turn, arrives through PD and LGD rather than as a new risk type, and a stylized carbon-price shock raised a sector loan book's expected loss 3.46-fold while showing that the safest borrowers on base-case PD can be the most exposed to the transition, a ranking that matters more than the headline figure given how much of the transmission rests on assumptions no history can test.

None of these methods is self-policing: independent validation of both discrimination and calibration, within a governance structure like the three lines of defense of Chapter 8, is what catches a miscalibrated model before it causes a loss. As in market risk, a risk number, whether a VaR figure or a credit score, is only as trustworthy as its assumptions, so sound practice combines several methods and accounts openly for where they disagree.

9.10 Suggested Exercises

9.1 A loan has EAD of \$500,000, PD of 5%, and LGD of 50%. Compute its expected loss, and explain how each of the three components would need to change to cut the expected loss in half.

9.2 Using the collateral-coverage regression in Section 9.2.2, compute the predicted recovery rate and implied LGD for a loan with a collateral coverage ratio of 1.50, and explain why extrapolating the fitted line this far beyond the observed data (which ranged from 0.40 to 1.20) should be treated cautiously.

9.3 Using the logistic regression coefficients in Section 9.2.6, compute the predicted probability of default and the corresponding credit score for an applicant with a debt-to-income ratio of 22% and 6 years of credit history.

9.4 Using the same bond as Section 9.2.3, recompute the stressed expected loss under a milder recession scenario with PD = 6% and LGD = 65%, and compare the resulting multiple of the base-case expected loss to the $3.11\times$ figure computed in the text.

9.5 Using Table 9.2, verify by hand that the Gini coefficient is 1.0, and explain what would happen to the computed AUC if applicant E's true outcome had instead been "No default" rather than "Default."

9.6 Using the transition matrix in Table 9.4, compute the four-year cumulative default probability for a Medium-rated borrower.

9.7 Using the Merton model from this chapter, compute the one-year default probability for a firm with asset value $V_0 = \$200$ million, debt of $D = \$120$ million due in one year, asset volatility $\sigma_V = 30\%$, and a risk-free rate of $r = 5\%$.

9.8 A portfolio of 200 identical loans each has EAD of \$2,000, PD of 4%, and LGD of 45%, with defaults assumed independent. Compute the portfolio's expected loss, the standard deviation of portfolio losses, and its economic capital at the 99% level.

9.9 A bank's average annual gross income over the past three years is \$350 million. Using the Basic Indicator Approach, compute its operational risk capital requirement, and explain one advantage

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and one disadvantage of tying operational risk capital to gross income rather than to a bank's actual operational loss history.

9.10 Explain why a credit model can have a high Gini coefficient while still being poorly calibrated, and describe a concrete lending decision that would go wrong as a result even though the model ranks applicants correctly.

9.11 Explain, using the Gaussian copula case vignette, why a mathematically sound correlation model can still contribute to a systemic mispricing of risk, and describe what a stress test of the correlation assumption itself, rather than of PD or LGD individually, might have revealed before 2008.

9.12 Explain why reject inference is fundamentally an incomplete fix for the selection bias described in Section 9.2.6, and describe a concrete way a lender's approval policy could change over time such that a credit model validated on last year's approved applicants performs noticeably worse on this year's.

9.13 Using the transition matrix in Table 9.4, explain in your own words why a mark-to-market credit risk model would assign a Medium-rated bond a lower value than an identically priced High-rated bond, even in a year when neither bond actually defaults.

9.14 Pick two of the challenges listed in this chapter and describe a concrete scenario, not already used in the text, in which that challenge could cause a credit risk management system to understate a firm's true risk.

9.15 Using Section 9.4.2's single-factor formula, recompute the 99% worst-case default rate for the 100-loan portfolio under an asset correlation of $\rho = 0.12$ instead of $\rho = 0.20$, and explain in your own words why a lower asset correlation produces a lower worst-case default rate for the same average PD.

9.16 Using this chapter's mark-to-market bond example, recompute $\mathbb{E}[V]$ and its standard deviation if the Medium-rated yield were 9% instead of 7% (holding the transition probabilities, coupon, and default recovery fixed), and explain why the expected value falls by less than the price gap between the two Medium-rated scenarios might suggest at first glance.

9.17 Using this chapter's Herfindahl-Hirschman Index example, compute the HHI for a portfolio with the same \$100,000 total exposure concentrated in just 4 borrowers of \$25,000 each, and explain why this portfolio's expected loss is identical to, but its concentration risk is much higher than, the 100-loan portfolio's.

9.18 Using this chapter's EAD example, compute the exposure at default for a \$1,000,000 committed credit line currently drawn to \$400,000, assuming a CCF of 60% instead of 50%, and explain why a higher CCF assumption increases a portfolio's expected loss even if PD and LGD are unchanged.

9.19 A one-year CDS on a different issuer trades at a spread of $s = 350$ basis points, and the assumed recovery rate is $R = 35\%$. Using Section 9.3.3's credit-triangle approximation, compute the implied hazard rate and the one-year default probability, and compare the result to a Merton-model estimate for the same firm if one were available.

9.20 Combine a stressed PD = 6% with an asset correlation of $\rho = 0.15$ for the 100-loan portfolio, and compute the resulting economic capital, comparing it to both the base-case (\$2,381) and the fully stressed (\$15,704) figures computed in the text.

9.21 Explain why the CDS-implied default probability approach requires no estimate of asset volatility, unlike the Merton model, and describe the specific circumstance under which a CDS-implied estimate would not be available for a given borrower.

9.22 Using this chapter's Merton-model inversion example, explain in your own words why the firm's implied equity volatility (67.03%) is so much higher than its assumed asset volatility (25%), and describe what would happen to that gap, wider or narrower, if the firm instead had much less debt outstanding relative to its asset value.

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9.23 Using the firm in Exercise 9.7 ($V_0 = \$200$ million, $D = \$120$ million, $\sigma_V = 30\%$, $r = 5\%$), compute its Merton-model equity value E_0 and implied equity volatility σ_E, and explain what two market-observable quantities a real analyst would feed into a numerical solver, without knowing V_0 or σ_V in advance, to recover this same result in practice.

9.24 Using Table 9.3, merge the two lowest debt-to-income bands into a single band covering everything below 25%, recompute that band's weight of evidence and the characteristic's information value, and explain why merging bins can never increase information value and should therefore be done only when monotonicity or bin size demands it.

9.25 Recompute Section 9.2.8's points spread between the safest and riskiest debt-to-income bands using PDO = 25 instead of 20, and confirm that although the spread in points changes, the implied odds ratio does not. Explain what this says about which parts of a scorecard's scale are substantive and which are presentational.

9.26 Repeat Table 9.5's lifetime expected credit loss calculation for a *High*-rated borrower instead of a Medium-rated one, using the same transition matrix, LGD, exposure, and discount rate. Verify the second year's marginal probability by hand, report the resulting Stage 1 and Stage 2 provisions, and explain why the ratio between them is *larger* for the higher-quality borrower even though every provision figure is smaller.

9.27 The chapter's discrete-time hazard example weights the benign, base, and adverse scenarios 30/50/20. Recompute the probability-weighted lifetime ECL under weights of 20/50/30 and 10/50/40, and express each as an excess over the base-scenario-only figure of \$181.79. Explain why that excess grows as weight shifts toward the adverse scenario, referring to the convexity of expected loss in the macroeconomic factor.

9.28 Repeat Table 9.1's carbon-price stress assuming every borrower can pass half of its carbon cost on to customers ($\pi = 0.5$). Compute each sector's c and stressed PD by hand, confirm the book's stressed expected loss, and explain why the utilities sector's stressed PD falls by proportionally more than the transportation sector's.

9.29 Using Equation (9.2) with no pass-through, recompute the fossil-heavy utilities' stressed PD at a carbon price of \$300 per tonne, the level the NGFS associates with a net-zero path by 2035, and explain why tripling the carbon price more than triples the stressed PD. Then describe a second, LGD channel through which the same shock would raise the oil and gas sector's expected loss, and why that channel is missing from the table.

9.30 For the swap of Section 9.5.1, compute $\mathrm{EE}(t)$ and $\mathrm{PFE}_{95\%}(t)$ at $t = 1$ and $t = 4$ using Equation (9.11), and recompute the CVA of Table 9.7 for a counterparty whose CDS trades at 350 basis points with an assumed recovery of 35%, holding the exposures and discount factors fixed. By what factor does the CVA change, and why is it not simply 350/200?

9.31 Explain, using Section 9.5.5's latent-variable model, why the wrong-way multiplier on expected exposure grows as the counterparty becomes *safer* (a lower default probability, so a more negative threshold k), holding the correlation fixed. What does this imply for a bank that sells large amounts of protection to, or trades large swaps with, highly rated counterparties whose defaults would coincide with extreme market moves?

9.32 A private-credit fund reports returns from quarterly model marks that adjust only partly to news. Model the reported return as $r^*_t = w\, r_t + (1 - w)\, r^*_{t-1}$, where r_t is the true, serially independent return with standard deviation σ and $0 < w \le 1$. (a) Show that $\mathrm{Var}(r^*_t) = w\,\sigma^2/(2 - w)$ and that the reported returns have first-order autocorrelation $1 - w$. (b) For $w = 0.4$, compute the ratio of reported to true volatility, and the Sharpe ratio the fund would report if its true Sharpe ratio were 0.50. (c) Using the private-credit discussion in Section 9.3.3, explain why a portfolio optimizer or a Vasicek-style capital model fed with these reported returns would allocate too much to private credit, and suggest one way to undo the smoothing before using the data.

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9.33 **Challenge (real data).** Download the UCI Statlog (German Credit Data) dataset (1,000 loan applicants, 20 features, a binary good/bad credit-risk label) directly from the UCI Machine Learning Repository. (a) Fit a logistic regression predicting default and report its AUC and Gini coefficient on a held-out test split, the same validation performed on the toy six-applicant model in Section 9.2.6. (b) Treating the fitted probabilities as an estimated PD and assuming a flat 60% LGD, compute the portfolio's expected loss and compare it to the realized default rate in your test split. (c) Before fitting anything, write a short data-cleaning note on at least two features whose categorical codes (for example, checking-account status) are not self-explanatory from the column names alone, the same undocumented-category-code problem Chapter 21's capstone case study encounters on a different real dataset entirely.

9.11 Further Reading

- Saunders and Allen (2010), *Credit Risk Management In and Out of the Financial Crisis*. Covers credit scoring, rating models, and portfolio credit risk with a strong emphasis on the lessons of the 2008 crisis referenced in this chapter's case vignette.
- Merton (1974), "On the Pricing of Corporate Debt: The Risk Structure of Interest Rates." The original paper establishing the structural credit model developed in Section 9.3.1.
- Gupton et al. (1997), *CreditMetrics—Technical Document*. J.P. Morgan's original technical document introducing the ratings-migration, mark-to-market portfolio credit risk framework discussed in this chapter's treatment of rating transition matrices.
- BCBS (2010), *Basel III: A Global Regulatory Framework for More Resilient Banks and Banking Systems*. The primary regulatory source for the capital adequacy concepts introduced in this chapter's discussion of economic and regulatory capital.
- Li (2000), "On Default Correlation: A Copula Function Approach." The original paper proposing the Gaussian copula model discussed in this chapter's case vignette.
- Salmon (2009), "Recipe for Disaster: The Formula That Killed Wall Street." A widely read journalistic account of the Gaussian copula's role in the 2008 financial crisis, providing helpful context for this chapter's case vignette.
- Vasicek (2002), "The Distribution of Loan Portfolio Value." The original source for the single-factor correlated-default model developed in Section 9.4.2, underlying the Basel IRB capital formulas.
- Hull (2022), *Options, Futures, and Other Derivatives*. Covers credit default swap pricing and the hazard-rate approximation used in Section 9.3.3's market-implied default probability discussion in considerably greater depth.
- Hand and Henley (1997), "Statistical Classification Methods in Consumer Credit Scoring: A Review." The standard survey of the scorecard tradition developed in Section 9.2.8, written before machine learning entered the field and therefore unusually clear about what the classical method was actually optimizing.
- Lou et al. (2012), "Intelligible Models for Classification and Regression." The paper behind the Explainable Boosting Machine, which fits by boosting the same additive form Section 9.2.9 shows a credit scorecard has always had.
- BCBS (2015), *Guidance on Credit Risk and Accounting for Expected Credit Losses*. The supervisory expectations governing the lifetime expected-loss models discussed in this chapter, including the scenario-weighting requirement whose convexity consequence the chapter computes.
- Bellotti and Crook (2009), "Credit Scoring with Macroeconomic Variables Using Survival Analysis." Develops the discrete-time hazard approach used in this chapter on real consumer credit data, including the macroeconomic conditioning that a fixed transition matrix cannot express.

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- Gregory (2020), *The xVA Challenge*. The standard practitioner text on counterparty credit risk, covering the exposure profiles, netting and collateral, wrong-way risk, and the full family of valuation adjustments introduced in Section 9.5.
- Huge and Savine (2020), "Differential Machine Learning: The Shape of Things to Come." Introduces differential machine learning, training neural-network pricing approximations on both simulated values and their adjoint-computed derivatives, the technique Section 9.5.6 connects to XVA computation.
- NGFS (2024), *NGFS Climate Scenarios for Central Banks and Supervisors: Phase V*. The reference climate scenarios used by most supervisory exercises, with the transition and physical-risk narratives behind the stress test of Section 9.2.4.
- Crouhy et al. (2014), *The Essentials of Risk Management*. A practitioner-oriented synthesis spanning market, credit, and operational risk, model validation, and risk governance, tying together the full breadth of topics covered in this chapter and Chapter 8 in a single volume.

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